The Supply-Side Effects of Healthcare Privatization: Evidence from Puerto Rico*

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June 18, 2026

Abstract

We estimate the effects of privatizing an entire publicly operated healthcare delivery system by studying a bundled policy that changed facility ownership, provider payment, and patient access at once. We do so by exploiting Puerto Rico's 1993 healthcare reform, which sold or leased its public clinics and hospitals to private providers and shifted to paying private managed-care organizations by capitation. This is, to our knowledge, the first causal evidence on privatizing an entire delivery system, as distinct from the single-facility privatizations studied in prior work. Since the policy was rolled out across all the counties of the island between 1994 and 2000, we rely on the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021) to estimate its effects on the healthcare labor market and patient health. We find that the reform contracted the local healthcare sector in per-one-thousand resident terms. Employment fell by 13.0 percent and Medicare-certified hospital beds by 7.8 percent relative to their pre-reform means. However, patient health did not deteriorate on average, and infant mortality fell by 13 to 27 percent of its pre-reform mean in the urban, more populous, and more healthcare-intensive counties. A back-of-the-envelope calculation implies that the fiscal savings to the public system, about $967 million in 2025 dollars over the four years after the reform, were enough to make the reform's net welfare positive before counting the health gains, which add a further $3.08 billion at the U.S. EPA value of a statistical life.

JEL classification: H51, I11, I18, J21, J45.

*I would like to thank all faculty members that helped me along the way with their feedback: Dr. Shantannu Khana, Dr. Mindy Marks, Dr. James Dana.

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1 Introduction

Over recent decades, governments have increasingly opted to deliver care through the private sector rather than providing it directly. Whether this lowers costs without sacrificing quality is a question economists have long debated (Shleifer, 1998). Some evidence suggests that privatization increases firm productivity (Megginson & Netter, 2001), yet little is known about its effects on consumers. The concern is most severe in healthcare, where contracts are inevitably incomplete (Grossman & Hart, 1986), quality is hard to verify (Arrow, 1963; Dranove, Kessler, McClellan, & Satterthwaite, 2003), and competition is limited (Cutler & Scott Morton, 2013; Gaynor, Ho, & Town, 2015). Under these conditions, Hart, Shleifer, and Vishny (1997) argue that private ownership strengthens the incentive to cut costs, which can decrease the quality of care (since no contract can specify that) and incentivize providers to avoid the patients they cannot profitably treat. The net effect on spending and health is therefore not determined by ownership alone. It rests on the joint design of three margins: who owns the delivery system, how providers are paid (Gruber, 2017), and who is granted access (Finkelstein, 2007). Because they pull in opposite directions, the net effect of moving them together, which is what privatizing a delivery system entails, cannot be inferred from research that studies one at a time. Yet that is what the literature offers, so how the margins interact and which dominates remains unknown. We fill this gap by asking how privatizing an entire delivery system affects the per-capita supply of care and population health by studying Puerto Rico's 1993 health reform, which changed all three margins at once.

Healthcare tends to represent a sizable expenditure for governments. In the United States, Medicare and Medicaid together claim more than a fifth of federal spending (Medicaid and CHIP Payment and Access Commission, 2025). Therefore, as a move to decrease operational costs, healthcare has experienced a privatization trend in the last few decades. In the USA, the government's share of US hospital capacity fell by about 42 percent between 1983 and 2019 (Duggan, Gupta, Jackson, & Templeton, 2023), and systems such as the Department of Veterans Affairs and England's NHS increasingly buy care from private providers (Congressional Budget Office, 2021; Congressional Research Service, 2025; Farmer, 2023; King's Fund, 2026; National Audit Office, 2026). How that care is paid for has also changed, from fee-for-service toward a per-capita basis (capitation), which now accounts for 54 percent of Medicare beneficiaries and $462 billion of federal Medicare spending (Kaiser Family Foundation, 2024a), as well as more than half of Medicaid benefit spending (Gruber, 2017; Kaiser Family Foundation, 2025; Medicaid and CHIP Payment and Access Commission, 2025; Shepard & Wallace, 2026). Yet paying through these plans has not lowered costs: Medicare pays them about 22 percent more than comparable enrollees would cost under fee-for-service (Medicare Payment Advisory Commission, 2024). Access depends on eligibility, and the rules for who qualifies determine both how many people a program covers and how much it spends. In this regard, Medicaid and CHIP reach 78 million people, and Medicaid spending alone neared $957 billion in fiscal year 2024, roughly two-thirds of it federal (Kaiser Family Foundation, 2026; Medicaid and CHIP Payment and Access Commission, 2025). Each is consequential on its own, yet whether privatizing any of them lowers public spending without harming patients remains an open question (Duggan & Hayford, 2013).

In an attempt to answer this, the literature has identified each margin on its own: ownership, payment, and access have each been studied where the others are held fixed, so we know what each does alone but not how they combine when a policy moves all three at once. On ownership, shifting delivery from public to private hands can worsen care. Chan, Card, and Taylor (2023) find that patients treated in Veterans Affairs hospitals survive at higher rates than those treated privately, and Goodair and Reeves (2022) link greater NHS outsourcing to higher treatable mortality, though that evidence is observational rather than causal. Yet in both the public healthcare programs are kept as alternatives to patients and are never themselves privatized. The closest evidence to a true transfer, and the nearest to our own setting, follows a single facility sold to private owners. Such is the setting that Duggan et al. (2023) study, where they find that privatizing a public hospital cuts labor and raises mortality among elderly patients while private owners select the more profitable patients.

Related work shows how acquisition by a larger firm changes a facility's behavior, where acquired dialysis facilities raise doses of highly reimbursed drugs, replace skilled nurses with lower-skilled technicians, and waitlist fewer patients for transplant, raising hospitalizations and mortality (Eliason, Heebsh, McDevitt, & Roberts, 2020), while acquired independent hospitals obtain higher negotiated prices and cut support-staff, capital, and financing costs with no improvement in measured quality (Andreyeva, Gupta, Ishitani, Sylwestrzak, & Ukert, 2024). But each facility is converted inside an otherwise intact market that absorbs the patients it rejects and workers it lays off, so these estimates are local to one facility and cannot show what happens when an entire system is converted at once, which is what the policy we study did. On payment, moving from fee-for-service to capitated managed-care plans while providers stay private has lowered utilization in some settings, with contested effects on spending and quality (Aizer, Currie, & Moretti, 2007; Chorniy, Currie, & Sonchak, 2018; Curto, Einav, Finkelstein, Levin, & Bhattacharya, 2019; Dranove, Ody, & Starc, 2021; Duggan, 2004; Duggan, Gruber, & Vabson, 2018; Marton, Yelowitz, & Talbert, 2014). Some of these settings randomly assign enrollees to plans that share the same providers, which separates the plan's effect from the providers' (Agafiev Macambira, Geruso, Lollo, Ndumele, & Wallace, 2026; Geruso, Layton, & Wallace, 2023; Wallace, 2023). Yet only the payer changes here, while ownership and access stay fixed, so these studies isolate payment alone. On access, extending insurance coverage raises the demand for care (Finkelstein, 2007), a margin studied apart from ownership and payment. No study observes an entire public delivery system being privatized, while payment and coverage shift with it. Because each margin has only ever been identified in isolation, these estimates do not add up to the effect of moving all three together.

We try to fill this gap by exploiting Puerto Rico's 1993 health reform, an understudied setting that moved all margins as part of a bundled policy. As part of this policy, the Commonwealth sold or leased its public clinics and hospitals to private providers, replaced salaried public healthcare workers with capitation paid to private managed-care organizations, and enrolled the medically indigent in managed-care coverage, moving all three margins together (Section 2.2). We study the reform across all 78 counties of the island, which transitioned in five annual cohorts between 1994 and 2000, with a staggered difference-in-differences design built on the not-yet-treated estimator of Callaway and Sant'Anna (2021) and administrative data on employment, hospital capacity, and vital statistics. There are two main benefits that come from this unique setting: minimization of cross-state spillovers and exogeneity of funding. The first one arises because Puerto Rico is an island, so the not-yet-treated counties carry none of the cross-border patient and provider flows that complicate mainland designs. Similarly, federal Medicaid financing is capped rather than matched (unlike in states within the United States), the resource changes we measure are not an effect of a state's incentive to draw down federal funds.

This paper makes three contributions. The first is to the literature on privatization and ownership, which has measured the effects of changing ownership one facility at a time. Duggan et al. (2023) study the conversion of a single hospital within an otherwise intact market, and Chan et al. (2023) compare government and private provision with the surrounding market held fixed; each isolates the ownership margin while payment and access stay in place. We provide, to our knowledge, the first causal evidence on privatizing an entire delivery system, in which ownership, payment, and access move together across every market at once. The second contribution is to the literature on Medicaid managed care and the payer side of privatization, which studies how moving enrollees into capitated plans changes spending, plan choice, and the outcomes patients experience (Curto et al., 2019; Dranove et al., 2021; Duggan et al., 2018). That work observes the payer and the patient but not the provider, and so is silent on what privatization does to the supply of care itself. We document that effect by showing the per-capita reallocation of healthcare labor and capacity. More specifically, we estimate that employment per thousand residents fell to 13.0 percent and saw no change in pay, so the system decreased inputs rather than decreasing their price. Because we observe the quantity of care rather than what the program paid for it, the result speaks to how privatized providers reallocate resources. The third contribution is to the study of how market structure shapes privatization's effects. The supply response was not uniform across the island, since it concentrated in the densely populated counties where the public system had been largest, a cross-market heterogeneity that neither single-facility settings, nor payer-side settings (which average over geography) can identify.

The paper proceeds as follows. Section 2 describes the reform and the pre-reform system, Section 3 the data, Section 4 the theoretical framework, Section 5 the empirical strategy, Section 6 the results, Section 7 the robustness analysis, and Section 8 concludes.

2 Setting

2.1 The Pre-Reform Public Delivery System

For most of the twentieth century the Commonwealth of Puerto Rico delivered healthcare through an integrated public system built on the regionalized principles of Arbona and Ramirez de Arellano (1978): universal access at the point of service, a four-tier referral hierarchy, and direct public employment of clinical staff. The base tier comprised 78 Centers of Diagnostic and Treatment (CDTs), one per county, providing primary, prenatal, dental, and pharmaceutical care through salaried government clinicians; above them sat five Area Hospitals, six Regional Hospitals, and the supra-tertiary Puerto Rico Medical Center in San Juan. The system was the dominant source of care: in 1990 it operated about 40 percent of the Commonwealth's hospital beds and served roughly 1.8 million residents, about 52 percent of the population (Instituto de Administración y Política de Salud, 1995).

The reform responded to a long-standing two-tier division: care for the medically indigent fell to the Department of Health, while the quality a resident otherwise received depended largely on the ability to pay privately (Commonwealth of Puerto Rico, 1993). Contemporaneous assessments documented quantity and quality gaps favoring private providers and higher recorded mortality among the publicly served population (World Health Organization & Pan American Health Organization, 1998), and the enabling legislation attributed the decrease of public-sector quality to inadequate budgets, the rising cost of medical technology, bureaucratic centralization, and political interference, gaps that reform efforts since 1967 had not closed (Commonwealth of Puerto Rico, 1993). Fiscal pressure compounded these concerns, since health spending had reached $2.5 billion, about 11 percent of gross domestic product, by 1990 (World Health Organization & Pan American Health Organization, 1998), while federal Medicaid support was capped rather than open-ended and far below the income-based formula applied to the fifty US states, leaving the Commonwealth to finance much of public provision from its own budget (Kaiser Family Foundation, 2024b; Medicaid and CHIP Payment and Access Commission, 2020). Against this background the reform sought to contract private insurers and providers to extend quality care to all residents, and to the medically indigent in particular, while containing costs (Commonwealth of Puerto Rico, 1993).

2.2 The 1993 Reform

The Puerto Rico Health Insurance Administration Act (Act 72 of September 7, 1993) created a single public purchaser of coverage for the medically indigent, the Administración de Seguros de Salud (ASES), and restructured delivery along all three margins at once (Commonwealth of Puerto Rico, 1993; World Health Organization & Pan American Health Organization, 1998).[1] Ownership moved from public to private, where the 78 CDTs (one per county) and the 11 Area and Regional Hospitals were sold or placed under long-term lease to private providers contracted through the new managed-care organizations, while the Commonwealth retained the supra-tertiary Medical Center (Instituto de Administración y

[1] Eligibility extended to residents with household incomes below 200 percent of the Puerto Rico poverty threshold, alongside the existing Medicaid-eligible population; the benefits package covered inpatient and outpatient hospital care, physician and ambulatory services, pharmacy, dental, mental-health and substance-abuse treatment, and preventive services, without exclusions for pre-existing conditions. Cost-sharing was $1 to $4 per visit and $0.50 to $3 per prescription, scaled by income (Centers for Medicare and Medicaid Services, 2000, 2001, 2003; Commonwealth of Puerto Rico, 1993; Instituto de Administración y Política de Salud, 1995; World Health Organization & Pan American Health Organization, 1998).

Política de Salud, 1995). Healthcare providers payment moved from Department of Health salaries to capitation paid by ASES-contracted insurers at rates negotiated bilaterally with providers (Commonwealth of Puerto Rico, 1993; World Health Organization & Pan American Health Organization, 1998). Access moved from open entitlement at public facilities to enrollment in a single managed-care network (designated by county of residence) with primary-care gatekeeping and income-scaled cost-sharing. The public system had been an open outside option for the privately insured and the main one for the uninsured. Therefore, its transfer to private providers removed that fallback [2].

ASES divided the Commonwealth into eight administrative health regions and awarded each to a single managed-care organization through competitive bidding, creating regional monopolies, and rolled the regions out between February 1994 and July 2000 (Brugueras Fabre, 2002; World Health Organization & Pan American Health Organization, 1998) (see Figure 1 for a map of the cohorts across all counties). The order ran from the sparsely populated and outer-island regions to the capital of San Juan, so treatment timing is correlated with county urbanization, the identification concern we address in Section 5. We group the eight transition dates into five annual cohorts by year of transition: 1994 (10 counties), 1995 (12), 1996 (30), 1998 (25), and 2000 (San Juan alone). Because San Juan holds the largest concentration of healthcare capacity, we report its influence separately and show that excluding it attenuates the supply-side magnitude (Section 7).

Two contemporaneous federal changes bear on identification. The capped financing predates the reform and is absorbed by county fixed effects. The federal Section 936 corporate-tax credit for manufacturing in Puerto Rico was phased out beginning in 1996, over the same years as the rollout and unevenly across counties by their manufacturing exposure; we treat it as a contamination concern and test the estimates against it directly (Section 7). Financing detail and the post-2002 policy changes such as *Mi Salud* and *Vital* (i.e. the successor programs) are reported in Appendix 10 (Centers for Medicare and Medicaid Services, 2026;

[2] Since we lack data on the size of the new provider networks, we cannot say whether enrollees gained access to more providers or fewer.

Pan American Health Organization, 2007).

3 Data

The analysis combines six data sources, all used at the county level and at annual frequency within 1990–2001[3] dataset, which is only used as of 1991). We measure the local healthcare sector two ways. Employment comes from the U.S. Bureau of Labor Statistics Quarterly Census of Employment and Wages (QCEW) (U.S. Bureau of Labor Statistics, 2025), which reports county-quarter employment, wages, and establishment counts by NAICS code; the healthcare sector is the union of ambulatory care (NAICS 621), hospitals (622), and nursing and residential care (623), and we report NAICS 621 separately because it maps most directly onto the privatized county clinic network. Because the QCEW counts establishments of every ownership type, transferring facilities from public to private does not by itself move workers into or out of the measure, since a change in employment is a change in the number of healthcare jobs, not a reclassification across ownership categories. Hospital capacity comes from the Centers for Medicare and Medicaid Services Provider of Services (POS) files (Centers for Medicare and Medicaid Services, 2025), aggregated to the county-year. The two measures see different parts of the system: the POS files identify only Medicare-certified providers and so exclude the 78 CDTs, whose primary care did not require certification. The privatized clinic layer that was the reform's main target therefore shows up in QCEW employment, not in the bed count, which corroborates the hospital tier.

The Puerto Rico Department of Health's Vital Statistics reports (Puerto Rico Department of Health, 2025) supply the patient-health outcomes (the crude birth rate, infant mortality, the low-birth-weight share, and the government and private hospital-birth shares), along with county population and the crude net migration rate; the Department issued these only as unstructured annual PDFs, which we digitized, reconciled across years, and assembled into the structured county-year panel used here. The worker-margin outcomes

[3]Except for the (Centers for Medicare and Medicaid Services, 2025)

(labor-force participation, the employment-to-population ratio, and unemployment) come from the monthly county labor-force survey of the Puerto Rico Department of Labor and Human Resources (Puerto Rico Department of Labor and Human Resources, 2025). Two further sources support the robustness checks: the National Center for Health Statistics National Vital Statistics System (NVSS) natality files (National Center for Health Statistics, 2025) corroborate the county birth outcomes with individual records, and the Behavioral Risk Factor Surveillance System (BRFSS) (Centers for Disease Control and Prevention, 2025) supplies the individual adult coverage reports we use to test the access first stage. Appendix 10 gives the construction details.

The Vital Statistics population series (the Commonwealth's official population count) is the denominator for all per-thousand-resident outcomes, since it updates annually over the rollout, so the per-capita outcomes measure supply against a growing rather than a contracting base and are not an effect of out-migration from treated counties (Table 1). Per-capita scaling places the labor, supply, and health outcomes aligned with the literature (Hackmann, Heining, Klimke, Polyakova, & Seibert, 2025; Wallace, 2023). The balanced QCEW healthcare panel covers 59 counties and the balanced CMS panel 40, while the Vital Statistics and labor-force panels cover all 78; the QCEW shortfall reflects the Bureau's suppression of county-industry cells with few establishments or employment in this industry, which tilts the balanced healthcare sample toward the more-populous counties. The supply-side estimates are therefore identified on a sample that already over-weights dense counties, so their concentration in the metropolitan areas should be read as consistent with both genuine heterogeneity and sample composition, a caveat we address in the Section 7.

Table 1 reports summary statistics, and Table 2 the cohort-specific pre-reform (1990–1993) means of population, public-system intensity, healthcare and placebo-industry employment, labor-market activity, demographic flows, and patient-health covariates, with one-way ANOVA p-values for the joint null of equal cohort means. The cohorts differ systematically in urban intensity, reflecting the rural-to-urban rollout (Section 2.2).

4 Theoretical Framework

The reform moved three margins at once (ownership, payment, and access), and they need not push per-capita healthcare employment in the same direction. We organize their predictions around the property-rights theory of the firm, in which the allocation of ownership governs the strength of cost-reducing incentives when quality is imperfectly contractible (Hart et al., 1997; Shleifer, 1998). Under private ownership the provider keeps what it saves and cannot be penalized for cutting quality on dimensions no contract covers, so its incentive to economize strengthens while its incentive to sustain noncontractible quality weakens; because labor is the largest variable input in care delivery, this falls first on staffing, and the ownership margin predicts $\Delta L^{\text{ownership}} \leq 0$. Capitation pushes the same way through the same logic: paid a fixed per-member rate, the provider becomes the residual claimant on what it does not spend and has a direct incentive to economize on the inputs it can adjust (Aizer et al., 2007; Duggan, 2004; Marton et al., 2014), giving $\Delta L^{\text{capitation}} \leq 0$ absent an offsetting rise in the capitated rate.

The implication for per-patient noncontractible quality q is two-sided and depends on the level from which the reform started. Where the public system had been adequately resourced, the same cost-cutting lowers quality under private provision, $\Delta q \leq 0$, most visibly where the reform's effect was largest; where it had instead been underinvesting, private provision can raise quality rather than lower it, as competition has been shown to do in public hospital systems (Propper, 2018). Puerto Rico's pre-reform system was the underinvesting case (Section 2.1), so the sign of Δq is an empirical question, and one whose answer also bears on how to value the inputs the reform removed, a point we return to in the welfare analysis (Section 6.4).

The access margin is ambiguous because its two parts push in opposite directions. Extending insurance coverage to the medically indigent raised the demand for insured care, which supply-stimulus logic predicts will expand the local healthcare labor market (Dillender, 2022; Finkelstein, 2007; Hackmann et al., 2025); routing enrollees through a single managed-care network with gatekeeping and cost-sharing (Section 2.2) works the other way, dampening utilization, so the net sign of ΔL^{access} is itself an empirical question. Summing the three margins gives the reform's net effect,

$$\Delta L^{\text{reform}} = \underbrace{\Delta L^{\text{ownership}}}_{\leq 0} + \underbrace{\Delta L^{\text{capitation}}}_{\leq 0} + \underbrace{\Delta L^{\text{access}}}_{\gtrless 0}. \tag{1}$$

Our estimate recovers the combined effect, not the contribution of each, because the margins interact: the incentive to economize under capitation is stronger when the provider also owns the facility.

The three forces dominate in different counties: the access force where coverage gains were largest (poorer, higher-unemployment counties), the contractionary forces where the privatized public sector was largest (denser ones). We read this cross-sectional contrast as suggestive only, since county types differ along the same urban gradient that set the rollout order (Section 2.2) and the subsamples are too small to test formally; it cannot attribute the effect to a single margin, though the concentration of the contraction in dense, high-capacity markets is itself a feature of the estimates we report.[4]

5 Empirical Strategy

The empirical analysis exploits the staggered rollout of the reform across five cohorts of counties ($\mathcal{G} = \{1994, 1995, 1996, 1998, 2000\}$) to identify the average effect of treatment on the treated (ATT). Let $Y_{i,t}$ denote an outcome measured for county i in year t, let $G_i \in \mathcal{G}$ denote the treatment cohort of county i, and let $D_{i,t} = \mathbf{1}\{t \geq G_i\}$ be the post-treatment indicator. Because every county in Puerto Rico is treated by 2000, the analysis has no never-treated comparison group; identification relies instead on *not-yet-treated* counties, units in later cohorts $g' > g$ not yet treated by year t. The estimand of interest

[4]Appendix 10 reports the formal derivations of the ownership and capitation predictions, the quality corollary, and the mobility margin.

is the group-time average treatment effect $\mathrm{ATT}(g,t) = \mathbb{E}[\,Y_{i,t}(g) - Y_{i,t}(\infty) \mid G_i = g\,]$ for $t \geq g$ (Callaway & Sant'Anna, 2021), where $Y_{i,t}(g)$ and $Y_{i,t}(\infty)$ are the potential outcomes under treatment-in-cohort-g and no-treatment respectively. Identification rests on three assumptions: (i) conditional pre-treatment parallel trends for the not-yet-treated comparison group, $\mathbb{E}[Y_{i,t}(\infty) - Y_{i,t-1}(\infty) \mid G_i = g] = \mathbb{E}[Y_{i,t}(\infty) - Y_{i,t-1}(\infty) \mid G_i > t]$; (ii) no anticipation, $Y_{i,t}(g) = Y_{i,t}(\infty)$ for $t < g$; and (iii) staggered, irreversible adoption. Under (i)–(iii), $\mathrm{ATT}(g,t)$ is identified by the doubly robust difference-in-differences estimator

$$\widehat{\mathrm{ATT}}(g,t) \;=\; \mathbb{E}\big[Y_{i,t} - Y_{i,g-1} \,\big|\, G_i = g\big] \;-\; \mathbb{E}\big[Y_{i,t} - Y_{i,g-1} \,\big|\, G_i > t\big]\,. \tag{2}$$

The regression tables report the simple weighted average of $\widehat{\mathrm{ATT}}(g,t)$ across cohorts and post-treatment periods, with weights proportional to the number of post-treatment county-year observations; the event-study figures report the dynamic aggregation $\widehat{\theta}_\ell$ for $\ell \in \{-4, \ldots, +4\}$ with the reference period $\ell = -1$. Inference uses the multiplier bootstrap of Callaway and Sant'Anna (2021) clustered at the county level with 1,000 replications.[5]

Because the estimand and the parallel-trends assumption are estimator-specific, Section 7 re-estimates every main outcome with three alternatives: the canonical two-way fixed-effects event study, the extended two-way fixed-effects estimator of Wooldridge (2025), and the imputation estimator of Borusyak, Jaravel, and Spiess (2024).

The validity of the design turns on pre-treatment trends, not on pre-treatment levels. Because earlier-treated counties are systematically more rural than later-treated ones, the rollout produces cohort imbalance on observable pre-treatment levels, but under Callaway and Sant'Anna (2021) such imbalance does not threaten identification so long as the not-yet-treated comparison group satisfies assumption (i). We assess that assumption directly through the event-study pre-trend coefficients reported alongside each main result in Section 6.

[5]See Appendix 10 for the dynamic-aggregation equation and the technical details of the doubly robust three-step procedure.

Three features of the setting bear on the identifying assumptions, and Section 7 probes each. The first is geography, and it works in the design's favor: as an isolated island, Puerto Rico generated no spillovers to healthcare markets beyond it, removing the cross-border confound that complicates mainland privatization studies. The residual concern is internal, in labor and patient flows across treated and not-yet-treated counties that would contaminate the comparison group assumption (i) relies on; we assess it by re-estimating on counties that share no border with a county treated in a different year. The second is the IRS Section 936 corporate-tax phase-out, which began in 1996 and is contemporaneous with the rollout; because it fell unevenly across counties by manufacturing exposure, it is a potential confound, and we condition on manufacturing employment to test whether it accounts for the estimates. The third is the leverage of San Juan as the sole 2000 cohort: with no county treated later, its group-time effect has no not-yet-treated comparison and is weakly identified, so we report its influence separately.

6 Results

6.1 First Stage and Supply Contraction

The reform's first stage operated cleanly on the ownership margin. Table 3 reports the estimates on the share of births delivered at government and at private hospitals, the cleanest available proxy for which ownership class operated the inpatient delivery system in each county-year. The government-hospital share fell by 14.79 percentage points ($p < 0.01$), or 23.6 percent of its 62.55 percent pre-reform mean, and the private-hospital share rose by an offsetting 15.19 percentage points ($p < 0.01$). The two estimates are statistically indistinguishable in absolute magnitude and opposite in sign: the reorganization of births is essentially a one-for-one substitution from public to private facilities.[6] The first-stage

[6]The two shares are not exact complements because a small tail of births, typically under one percent, is recorded as non-hospital or with unspecified ownership.

event study in Figure 1 confirms the timing: pre-treatment coefficients on both series are statistically indistinguishable from zero through $\ell = -1$, and the substitution opens within the first year of treatment and widens through $\ell = +3$.

Coincident with the ownership transition, per-capita healthcare employment fell. Table 4 reports the QCEW labor-market estimates for the healthcare aggregate (NAICS 621+622+623) and, separately, for ambulatory primary care (NAICS 621), the subsector mapping most directly onto the privatized county clinic network. Healthcare employment per thousand residents fell by 0.511 employees, 13.0 percent of its pre-treatment mean ($p < 0.01$), with a larger decline of 17.3 percent in ambulatory primary care ($p < 0.01$). Establishments per thousand fell by 6.6 percent in the aggregate ($p < 0.01$) and by 4.6 percent in NAICS 621 ($p < 0.10$), so the adjustment operated on both the intensive employment margin and the extensive firm-count margin.

This per-capita decline is a level result, and it does not mean the sector shrank everywhere. Appendix Table A.7 reports the same outcomes in a $\log(1+x)$ specification alongside the per-capita levels, and the two diverge in sign: per thousand residents, employment falls 13.0 percent and establishments 6.6 percent, while in logs employment rises 19.0 percent and establishments 14.4 percent. The contrast reflects heterogeneity in where the adjustment fell. The per-capita level estimates are dominated by the populous counties and turn negative because the large urban delivery systems contracted in absolute terms, while the log specification is positive, consistent with proportional expansion among the smaller, lower-baseline rural counties. The reform reallocated healthcare activity away from the dense urban centers rather than contracting it uniformly, and the per-capita level estimate is correspondingly sensitive to the densest counties, attenuating when San Juan is excluded (Section 7).

The adjustment fell on the quantity of labor, not its price. The wage-per-employee estimate is statistically indistinguishable from zero in both panels: employment and establishment counts dropped while real compensation per worker held flat. This is what the ownership and capitation mechanisms predict (Section 4): providers economize on the largest variable input rather than cut wages. Duggan et al. (2023) find that privatized U.S. hospitals cut labor intensity by about 8 percent with no significant change in compensation, and the same input economizing appears under Medicaid capitation (Duggan, 2004). Private payers show the same quantity-not-price pattern on the demand side: Medicare Advantage spends 9 to 30 percent less per enrollee than traditional Medicare through lower utilization rather than lower prices (Curto et al., 2019), and privatizing the Medicaid drug benefit cut drug spending by about 21 percent through cheaper-product substitution (Dranove et al., 2021). Those estimates describe a private insurer economizing on the care it pays for; ours describes the delivery system itself shrinking the inputs it operates. Figure 2 reports the corresponding six-panel event study for employment, establishments, and the wage bill per employee across both industry definitions: pre-treatment coefficients are statistically flat through $\ell = -1$, and the post-reform contraction in employment and establishments opens within the first year while the wage-per-employee series remains flat throughout.

The establishment decline is also a change in market structure, not only in inputs. Fewer establishments per resident means fewer providers competing in each market, most of all in the dense urban centers where the contraction concentrated. A literature on hospital competition finds that this matters for care: in the English NHS, where hospitals compete on quality at administered prices, greater competition has been linked to lower mortality and better management (Bloom, Propper, Seiler, & Van Reenen, 2015; Cooper, Gibbons, Jones, & McGuire, 2011; Gaynor, Moreno-Serra, & Propper, 2013). Reduced competition therefore gives the reform a second route to quality, separate from the cut in inputs, and one a single-facility privatization cannot register, because there the surrounding market stays intact and the number of competitors is unchanged; only converting a whole system at once thins the field of providers. The net effect on quality is ambiguous: the loss of competition pushes it down, while the replacement of an underinvesting public monopoly with private provision pushes it up (Section 4), and we take up the realized effect with the patient-health estimates (Section 6.3).

Physical hospital capacity contracted as well, on the narrower CMS panel of forty Medicare-certified counties. These estimates appear alongside the patient-health outcomes in Table 5, Columns 3–4: Medicare-certified hospital beds per thousand fell by 7.8 percent ($p < 0.01$) and Medicare-certified services per thousand by 4.9 percent ($p < 0.05$). The contraction therefore spans both layers of the delivery system, the ambulatory clinic network (QCEW employment and establishments) and the inpatient hospital infrastructure (CMS beds and services), though the bed estimate shares the per-capita level result's sensitivity to the densest counties (Section 7). The bed and service event studies appear in Figure 3, Panels (e)–(f), where the post-treatment contraction opens by $\ell = +2$.[7]

6.2 Worker-Side Adjustment Margins

The labor-market contraction did not show up in wages or in the unemployment rate. Table 6 reports the margins along which displaced healthcare workers and broader local-labor-market spillovers might adjust. The labor-force-participation rate fell by 1.11 percentage points, 3.5 percent of its 31.895 percent pre-treatment mean ($p < 0.01$), while the BLS-standard unemployment rate is statistically indistinguishable from zero against a pre-reform mean of 17.770 percent.[8] The crude net migration rate moves from a pre-treatment mean of +1.06 residents per thousand to an estimated post-treatment −14.65 per thousand ($p < 0.05$), and the crude birth rate falls 3.1 percent (−0.564 per thousand, $p < 0.05$). Total county population, by contrast, is not lower in treated units: the population-level estimate is +5.1 percent ($p < 0.01$).[9] These estimates rule out a population exodus as the driver

[7] The supply magnitudes reported here are simple-aggregate ATTs, averaged over all post-treatment periods. The event studies in Figures 2 and 3 show the per-capita contraction deepening with time since treatment, so the longest-horizon coefficients exceed these averages, with correspondingly wide confidence intervals. We report the simple-aggregate ATT throughout and carry it, rather than the larger long-horizon coefficients, into the welfare accounting (Section 6.4) as the conservative choice.

[8] The unemployment rate is reported not in Table 6 but in Online Appendix Table OA.1, alongside the labor-force-participation and employment-to-population rates; Appendix 10 reports the participation event study.

[9] Online Appendix Tables OA.20 and OA.21 report the population estimate in levels and logs and an explicit demographic-balancing check. The population-level and net-migration estimates do not arithmetically reconcile, because the migration rate is a residual of three separately measured flows (Equation 4), more

of the contraction: treated counties grew even as participation and the crude birth rate fell. We do not observe age composition and so infer no specific demographic recomposition; the participation and fertility declines establish only that these margins moved where the population did not contract. The net-migration and crude-birth-rate event studies appear in Figure 3, Panels (c)–(d), the migration signal opening within two years of treatment and the birth-rate trajectory bending downward from $\ell = 0$.

These margins also bound what we can say about the displaced workers themselves. Because unemployment did not rise, they were not absorbed into local joblessness; the adjustment fell instead on participation and migration. County aggregates cannot follow individuals, so we cannot separate labor-force exit, migration off the island, and reallocation into other local sectors. Two features make selective provider emigration the more plausible reading: the population grew, so any migration was selective rather than broad, and Puerto Rico's clinicians hold U.S. credentials that require no re-licensing on the mainland (Appendix 10), lowering the cost of leaving for precisely the workers the reform displaced. The negative net-migration estimate is consistent with such selective outflow, though the aggregates cannot confirm it.

The net migration rate is the one primary outcome whose sign is sensitive to estimator choice, so we read its magnitude as suggestive (Section 7). The participation result, on the same 78-county panel and the behavioral counterpart of the migration story, is robust across all four estimators and is the firmer basis for the labor-supply reading.

6.3 Patient Health, Heterogeneity, and Access

The short-horizon patient-health response is null on average. Table 5, Columns 1–2, reports the county-aggregate Vital Statistics outcomes whose biological response horizons fall within the four-year post-reform window: the infant mortality rate fell by 1.205 deaths per thousand live births (9.5 percent of its pre-mean, statistically insignificant) and the low-birth-weight

sensitive to compounding measurement error than the level outcome, and because the C&S aggregation weights differ across panels of different length.

share rose by 0.06 percentage points (statistically insignificant), and both event studies in Figure 3, Panels (a)–(b), are flat in the pre- and post-treatment windows.[10]

The cause-specific mortality estimates in Online Appendix Table OA.25 refine this average null into a set of offsetting movements across two groups that proxy different dimensions of care. The first collects deaths that timely preventive and chronic-disease care can avert, and on the cardiovascular margins it points to improvement: heart-disease mortality falls 6.1 percent ($p < 0.05$) and cerebrovascular mortality 11.0 percent ($p < 0.05$), while diabetes and pneumonia or influenza mortality are unchanged. Maternal mortality moves the other way, rising 73 percent of its small pre-reform base (+0.002 deaths per thousand residents, $p < 0.10$); because a maternal death usually reflects prenatal risk left unmanaged rather than delivery-room failure, this is consistent with acute delivery capacity holding while the preventive care that depends on sustained patient contact thinned. The second group, deaths from accidents and homicide, proxies emergency and trauma care, where survival turns on acute-care access that often crosses county lines; both rise, accident mortality by 40 percent (statistically insignificant) and homicide mortality by 75 percent ($p < 0.01$), consistent with the contraction of beds and facilities lengthening the path to emergency care. We read the homicide estimate cautiously, because its recorded count blends the incidence of violence, which rose across Puerto Rico through the 1990s for reasons unrelated to the health system, with the case-fatality rate emergency care governs, and the design cannot separate the two. These acute-care deaths are readable within the four-year window through case-fatality, where chronic-disease mortality is not, and the Hispanic health paradox (Fernández, García-Pérez, & Orozco-Alemán, 2023) implies measured health can deteriorate well before mortality responds. Across the within-window outcomes, then, the privatized system delivered fewer per-capita inputs with offsetting movements in measured health rather than a broad deterioration: improvement on the cardiovascular margins, no average change in infant mortality, and deterioration on the obstetric margin and, more tentatively, on emergency-care access.

[10] Individual-level NVSS specifications on birth weight, prenatal-visit count, trimester of prenatal-care initiation, and Apgar scores are similarly null (Online Appendix Table OA.22).

Table 7 partitions the 78 counties at the median pre-reform government-hospital birth share, and the first-stage substitution is larger in the below-median group (-15.49 percentage points, $p < 0.01$) than the above-median group ($+2.41$ percentage points, statistically insignificant). The baseline share is the wrong lens on where the reform bit, because pre-reform reliance on public hospitals was highest in the rural, poorer counties with the fewest private alternatives. That high-reliance half is largely the rural periphery, where the same facilities went on delivering most births, so measured substitution was small; the realized ownership transition instead concentrated in the urban, more populous, higher-capacity, richer, and lower-unemployment counties. Partitioning on pre-reform population density, population, healthcare employment, the poverty rate, or the unemployment rate, the government-hospital share falls by 24 to 31 percent in the urban, larger, higher-capacity, richer, and lower-unemployment half and is flat in the other (Appendix Table A.6 and Online Appendix Tables OA.9, OA.10, OA.23, and OA.24).

The infant-mortality response follows the same gradient. In the halves where the ownership transition took hold, infant mortality falls by 13 to 27 percent of its pre-treatment mean, statistically significant across the density, population, healthcare-employment, poverty, unemployment, and commuting partitions; in the halves where the substitution was muted it does not improve, and in the smallest counties it rises, a marginally significant 28 percent.[11] The quality gain follows the realized ownership transition, concentrated in the urban counties with the private infrastructure to absorb the transfer, consistent with the gains the framework predicts for under-resourced public systems (Section 4). These counties differ on many dimensions at once and the subsamples are small, so we read the pattern as consistent with quality gains from the transition rather than a clean isolation of which margin produced

[11] For the half in which the ownership transition concentrated, the infant-mortality ATTs in deaths per thousand live births are -2.03 (urban, $p < 0.05$), -2.56 (larger population, $p < 0.01$), -2.63 (higher healthcare employment, $p < 0.01$), -1.62 (lower poverty, $p < 0.10$), -2.62 (lower unemployment, $p < 0.05$), and -3.12 (workplace-center, $p < 0.01$); the government-hospital substitution in these halves ranges from -14.2 to -18.2 percentage points. In the opposite halves the infant-mortality estimates are positive and insignificant, except the smallest-population half at $+3.66$ ($p < 0.10$). Appendix Table A.6 and Online Appendix Tables OA.9, OA.10, OA.23, OA.24, and OA.11 report all six partitions.

them. The improvement does not concentrate in the poorest counties, where the coverage expansion delivered its largest gains, so it does not trace the access margin, and it is hard to square with the uniform quality deterioration a Hart et al. (1997) reading predicts. It also runs opposite to Duggan et al. (2023), who find that U.S. hospital privatization raised elderly mortality: the divergence is largely population and horizon, since their effects fall on a horizon our four-year perinatal window does not reach, while among the adult causes we can read at a short horizon, maternal mortality moves up as their elderly mortality does and cardiovascular mortality moves down. The payer-side evidence fits the no-harm side: Duggan et al. (2018) find that Medicare Advantage restricted hospital access with no measurable mortality effect, and Chorniy et al. (2018) find that Medicaid managed care raised screening but with more emergency-room use and no clear health gain, both comparing a private payer to a public one rather than private to public delivery, so they bound our interpretation rather than rival its magnitude. The supporting heterogeneity event studies, by baseline poverty rate and by pre-reform government-hospital exposure, appear in Figures 4 and 5, each splitting the panel at the median and tracing employment, infant mortality, and net migration for the two halves.

The access margin moved in the expansionary direction. Across age, education, and employment subgroups, self-reported insurance coverage in the BRFSS rises by 3 to 10 percentage points, estimated under the Wooldridge (2025) extended two-way fixed-effects estimator because the post-1996 BRFSS window admits only the later cohorts.[12] This is the opposite of what facility-level privatization produces in the United States: Duggan et al. (2023) find that private providers raise revenue by cream-skimming profitable patients and cut admissions of low-income patients at the market level. The Puerto Rico reform was instead built as a coverage expansion for the medically indigent, so its enrollees were the

[12] Online Appendix Tables OA.14–OA.19 report the six BRFSS subgroup estimates; the "has health plan" point estimates range from +2.8 percentage points (ages 65 and older, where near-universal Medicare leaves little room to expand) to +10.5 percentage points (ages 45–64), and are significant in five of the six subgroups. Appendix 10 discusses the subgroup pattern and the single-cohort-exposure limitation that requires the ETWFE estimator on this panel.

low-income group itself, and access on the insurance margin widened rather than narrowed. This coverage expansion is the access margin's empirical counterpart; combined with the supply contraction documented above, it shows the contractionary equilibrium held even as coverage expanded, the configuration in which the supply-stimulus prediction of Finkelstein (2007), Dillender (2022), and Hackmann et al. (2025) reverses sign once coverage is bundled with delivery-system privatization, capitation, and a single managed-care network.

6.4 Welfare Bounds

We close by tying the estimated supply and health effects into a cost-benefit calculation, following Duggan et al. (2023), to make the policy trade-off concrete. It captures only the margins we measure, omitting health effects beyond the four-year horizon, the value of the coverage expansion to enrollees, and other margins a complete analysis would require; it is a back-of-the-envelope, partial-equilibrium calculation in real 1982–1984 dollars, read for sign and order of magnitude rather than as a point estimate. Table A.4 reports every cell in both price bases.

Net welfare is the avoided cost of the healthcare inputs the reform no longer purchased plus the value of any averted infant deaths:

$$W = \underbrace{\frac{P\,T}{1000}\Big(|\hat{\beta}_{\text{bed}}|\,c_b + |\hat{\beta}_{\text{emp}}|\,\bar{w}\Big)}_{\text{avoided cost}} + \underbrace{v\,\frac{|\hat{\beta}_{\text{IMR}}|}{1000}\,B_H\,T}_{\text{value of averted infant deaths}} . \tag{3}$$

We depart from Duggan et al. (2023) in three respects, each for a reason. First, the cost saving is the avoided cost of the inputs the delivery system shed, the per-thousand reductions of $\hat{\beta}_{\text{bed}} = -0.265$ beds (Table 5) and $\hat{\beta}_{\text{emp}} = -0.511$ workers (Table 4) scaled to the island population over the four post-reform years, rather than the hospital deficit they observe and net against tax revenue; we measure the physical inputs directly, and unlike the discrete hospital privatizations they study, the health reform privatized an entire delivery system whose budget we do not observe at the provider level. A bed-year is priced at its implied total operating cost c_b from the CMS cost reports and a worker-year at the 1990–1993 mean healthcare wage $\bar{w}$. Second, the health term is a benefit rather than a cost, since infant mortality fell where they find added deaths among the elderly. Third, because the full-panel infant-mortality reduction is statistically insignificant, we value it only as an upper bound, applying it over the births in the high-substitution half B_H of Section 6.3, one-half the island total, at the U.S. EPA reference value of $10 million per statistical life (U.S. Environmental Protection Agency, 2010; Viscusi & Aldy, 2003), or $3.19 million in 1982–1984 dollars.

Under the zero-health choice the bound rests on the avoided cost alone and is $300.8 million over the window under the headline employment estimate ($21.5 per resident per year). Conditioning the employment reduction on the contemporaneous manufacturing-tax phase-out (Section 7) attenuates it to -0.224 per thousand and lowers the avoided cost to $262.5 million ($18.7 per resident per year), our most conservative figure. Adding the value of the averted infant deaths, $958.7 million, raises net welfare to $1,259.5 million under the headline supply estimate and $1,221.2 million under the §936-conditional one ($90.0 and $87.2 per resident per year, or roughly $4.0 and $3.9 billion in 2025 dollars).

We take the zero-health cells as our central estimate: they are positive, rest only on the avoided cost, and do not depend on the patient-health margin in either direction; among them we prefer the §936-conditional figure, which nets out the manufacturing-tax phase-out. The health-valued cells are an upper bound, for three reasons. The infant-mortality benefit they monetize is insignificant on the full panel and recovered only in subgroups; they price that benefit while ignoring the adult cause-specific effects, which move both ways, the favorable cardiovascular declines against the adverse maternal increase, the clearest identified short-horizon harm; and they apply an adult statistical-life value to infant deaths, understating the per-death value. Three further margins we cannot quantify, none changing the sign: mortality beyond the four-year window, the value of the coverage expansion to enrollees, which low-income willingness-to-pay places below its cost (Finkelstein, Hendren, & Shepard, 2019), and the share of the saving that reflects reduced provider rents rather than freed real resources, since much healthcare spending is such a transfer (Finkelstein, Hendren, & Luttmer, 2019).

7 Robustness

This section reports six robustness exercises, each varying one input to the main specification (the estimator, the sample, the placebo industry, the conditioning covariate, or the outcome normalization) while holding the others at their main-text defaults; Figure **??** is a forest-plot summary for healthcare employment per 1,000, and the Appendix tables report the full specifications for every primary outcome. One pattern runs through all six. The first-stage ownership transition is robust everywhere and the patient-health outcomes are robust; the per-capita supply and labor-market contractions hold across the alternative estimators and survive the spillover restriction as a loss of power, but attenuate to insignificance when the metropolitan capital is dropped and when the estimate is conditioned on pre-reform manufacturing exposure. We therefore read the supply-side magnitude as the population-weighted effect on residents, concentrated in the metropolitan core where most of the population and the public delivery system were located, rather than as a uniform islandwide contraction; the manufacturing-conditional estimate provides the conservative bound we carry into the welfare accounting.

Estimator class. We triangulate the main estimates against three alternatives: the canonical two-way fixed-effects (TWFE) event study, the extended two-way fixed-effects estimator of Wooldridge (2025), and the imputation estimator of Borusyak et al. (2024). TWFE serves as a contamination diagnostic: under treatment-effect heterogeneity its event-time coefficients are negatively weighted averages prone to sign reversals that the doubly robust C&S, ETWFE, and BJS estimators avoid. ETWFE, C&S, and BJS agree in sign and magnitude on the first-stage transition, the healthcare-employment contraction, the labor-force-participation decline, and the hospital-bed contraction, while TWFE produces the expected sign reversals on the outcomes where heterogeneity is most pronounced (healthcare employment, infant mortality, and the low-birth-weight share).[13] The crude net migration rate is the one outcome on which the four do not agree: TWFE, ETWFE, and BJS return insignificant estimates of mixed sign (-5.69, -1.53, and $+4.51$ per thousand), while the C&S estimate of -14.65 per thousand ($p < 0.05$) is the basis for the migration discussion in Section 6.2, so that result is suggestive: its sign is the C&S signature but its magnitude is not robust.

Excluding San Juan. San Juan is the sole 2000-cohort county and holds roughly 11 percent of the Commonwealth's population. Dropping it attenuates the supply-side and labor-market estimates substantially while leaving the first stage intact: the government-hospital-share estimate moves from -14.79 to -8.22 percentage points (both $p < 0.01$),[14] healthcare employment from -13.0 to -5.3 percent of pre-mean and the hospital-bed estimate from -7.8 to $+3.0$ percent, the latter two no longer significant. The attenuation is mechanical: the metropolitan capital holds the largest concentration of healthcare workforce and Medicare-certified capacity, so dropping it strips out most of the absolute mass through which a per-capita contraction is identified, and it is also the cell where the out-migration option is cheapest in the legal-portability sense (Section 4). The first-stage transition depends on neither feature and survives.

Within-island spillovers. The not-yet-treated comparison requires that already-treated neighbors not affect comparison counties (Section 5). Two restrictions tighten this at the cost of power: a non-border subset dropping counties that share a boundary with a different-

[13] Appendix Table A.1 reports the four-estimator point estimates and bootstrap standard errors for the six primary outcomes, and the full TWFE specification (county and year fixed effects with event-time indicators, reference period omitted, standard errors clustered at the county level). Across estimators, the first-stage government-hospital-share estimate is -12.9 percent (TWFE), -26.4 percent (ETWFE), -23.6 percent (C&S), and -18.0 percent (BJS) of pre-mean, all $p < 0.01$; HC employment per thousand is -25.7 (ETWFE), -13.0 (C&S), and -16.7 (BJS), all $p < 0.01$; LFP per population is -3.6, -3.5, and -2.2 percent, all $p < 0.01$; hospital beds per thousand is -13.1, -7.8, and -5.6 percent, all $p < 0.01$.

[14] Online Appendix Table OA.3 reports the first-stage and patient-health side-by-side specifications; Appendix Table A.2 reports the QCEW labor-market and CMS hospital-capacity specifications; Online Appendix Table OA.4 reports the ownership-stratified CMS robustness.

cohort treated unit, and a low-commuting-inflow subset dropping the upper half of the workplace in-commuting distribution.[15] The first-stage transition is robust under both (-12.14 percentage points on the non-border subset, -7.18 on the low-commuting subset, both $p < 0.05$), as are the patient-health outcomes. The labor-market estimates do not survive the non-border restriction at its sample size of 27 QCEW counties, where healthcare employment, the wage bill per employee, and establishments are each indistinguishable from zero, and the hospital-bed and services estimates keep the main direction but lose precision. The cleanest reading is power loss rather than evidence against the main estimates.

Placebo industries. These specifications test whether contemporaneous shocks orthogonal to the reform, most prominently the Section 936 corporate-tax phase-out beginning in 1996, produce healthcare-shaped effects on industries with no exposure to the privatized delivery system: NAICS 11 (agriculture), NAICS 48–49 (transportation), and NAICS 71 (arts). NAICS 31–33 manufacturing is excluded *a priori*, since the Section 936 phase-out shocks it during the window, making it endogenous to the same policy environment as healthcare rather than orthogonal.[16] The evidence is mixed. Agriculture is null on absolute-headcount and per-capita employment and establishments, its per-capita wage-bill decline not surviving raw-level normalization (population redistribution, not an agriculture-specific shock). Transportation contracts in both per-capita and raw-level form beyond what redistribution can produce, consistent with input-output linkage to the phasing-out manufacturing logistics chain through Section 936 sub-industries. Arts is underpowered ($n = 9$). The transportation failure through a known confounder motivates the direct manufacturing-control test next.

[15] Appendix Table A.5 and Online Appendix Tables OA.6, OA.5, OA.7, and OA.8 report the corresponding specifications. The non-border restriction retains 27 of 78 counties on the QCEW panel, 19 of 40 on the CMS panel, and 39 of 78 on the Vital Statistics panel; the low-commuting-inflow restriction retains 64 of 78 on the Vital Statistics panel and 34 of 40 on the CMS panel.

[16] Online Appendix Tables OA.12 and OA.13 report the per-thousand-resident and raw-level placebo specifications.

Manufacturing control. The transportation placebo failure motivates a direct test of whether the main contraction is identified in isolation from the Section 936 phase-out. We condition the main healthcare-employment estimate on a pre-reform manufacturing-exposure covariate, the 1990–1993 mean of NAICS 31–33 employment per thousand; using a strictly pre-1996 baseline rather than the contemporaneous series ensures the covariate is not itself shocked by the policy whose contamination the test nets out.[17] The C&S estimate attenuates from −14.7 percent of pre-mean (no control, $p < 0.01$) to −6.8 percent (with the covariate, statistically insignificant), remaining negative and not driven by sample restriction. The result is therefore neither pure §936 contamination, since the conditional estimate is negative rather than zero, nor identified in isolation at full magnitude. The supportable reading is intermediate: part of the cross-sectional heterogeneity in the contraction loads onto pre-reform manufacturing exposure, a correlation that may reflect §936 contamination but may also reflect that high-manufacturing counties are urban and industrial and so overlap with the dimensions on which the contraction concentrates (Section 6.3). The interpretation rests on the conjunction of the policy-tied first stage, which this conditioning leaves intact, the heterogeneity-as-mechanism evidence, and the negative conditional estimate.

Per-capita normalization. The per-capita normalization is operative throughout, for the welfare-relevance and comparability reasons given in Section 3. The −13.0 percent contraction in healthcare employment per thousand coexists with roughly +19.0 percent growth in raw-level $\log(1 + y)$ employment over the same window: the local healthcare sector grew in absolute terms but more slowly than the local population.[18] The two are not in conflict, each identifying a distinct object: the per-capita estimate the change in

[17] Appendix Table A.3 reports the conditioning specification. The sample is the 54 counties with non-missing 1990–1993 baseline manufacturing data, a tightening of the 78-county main-text panel imposed by the conditioning covariate. The C&S no-control estimate on this sample is −14.7 percent of pre-mean, economically equivalent to the 78-county main result of −13.0 percent reported in Table 4, so the sample restriction itself does not drive the conditioning result.

[18] Appendix Table A.7 reports per-1,000 and $\log(1 + y)$ specifications for QCEW healthcare employment, establishments, and the wage bill side by side; Online Appendix Table OA.2 reports the parallel specification on $\log(1 + y)$ per-1,000 normalizations and $\log(1 + y)$ raw-level outcomes.

resources per resident, the welfare-relevant scaling, and the raw-level estimate the change in absolute headcount, mechanically larger wherever the denominator is itself changing. The distributional finding of the paper rests on the per-capita denominator, not the absolute headcount.

8 Conclusion

Puerto Rico's 1993 health reform sold or leased its public diagnostic and treatment centers and Area and Regional Hospitals to private providers and shifted to paying private managed-care organizations prospective capitation through the Commonwealth's new Health Insurance Administration, retaining only the supra-tertiary referral complex. The first stage confirms the ownership transfer, and the post-reform equilibrium delivered fewer healthcare resources per capita: healthcare employment, ambulatory primary-care employment, the establishment count, and Medicare-certified hospital beds and services all contracted relative to population. The contraction loaded onto labor-force participation rather than unemployment, alongside declines in the crude birth rate and net migration; the displaced workers were not absorbed into local unemployment, and the participation and migration declines are consistent with labor-force exit, with out-migration made feasible by U.S. citizenship and portable medical credentials, or with both, which the aggregate data cannot distinguish. Measured patient health did not broadly deteriorate within the four-year window and improved where the ownership transition took hold, infant mortality falling 13 to 27 percent of its pre-mean in the urban, more populous, higher-capacity, and richer counties.

The main finding is that a system-wide delivery-system privatization can deliver less measured healthcare per capita without a proportional loss in measured patient quality. The estimates do not vindicate the Hart et al. (1997) quality-deterioration corollary in this setting, since infant mortality fell where the reform had the most bite, nor the supply-stimulus prediction of the insurance-expansion literature (Dillender, 2022; Finkelstein, 2007;

Hackmann et al., 2025) in its unconditional form, since the local healthcare labor market contracted rather than expanded under the bundled treatment of capitation, single-network confinement, and ownership transfer. They support a more modest claim: an ownership-and-payment reform that constrains the per-enrollee budget can reduce per-capita inputs without proportional welfare losses when the pre-reform public system was underinvesting on the noncontractible quality margin (Hart et al., 1997; Propper, 2018). A back-of-the-envelope bound places the short-horizon welfare consequence between \$844 million and \$4.0 billion in 2025 dollars over the four-year window (Section 6.4), with the avoided cost the binding source of welfare improvement even when patient health is valued at zero.

For the privatization debates the paper opened with, the contribution is the counterfactual itself. The reforms that animate those debates each move a partial bundle of the three margins Puerto Rico moved together: the Department of Veterans Affairs MISSION Act added outside-option access without transferring ownership, the English internal market introduced competition without capitating public delivery, and risk-based Medicaid managed care capitates payment without transferring facilities. Puerto Rico is the case in which the bundle is complete and the delivery system itself changes hands, so it speaks to the public-versus-private-delivery question the payer-side and single-facility literatures cannot reach: those designs change who pays while providers stay private (Chorniy et al., 2018; Curto et al., 2019; Dranove et al., 2021; Duggan et al., 2018; Wallace, 2023), or compare public and private provision (Chan et al., 2023), or convert a single facility within an intact market (Duggan et al., 2023); none observes the same delivery system run publicly and then privately. Puerto Rico did so through procurement to a single plan per region, the market design Cuesta and Tebaldi (2025) show can raise welfare relative to regulated competition by removing cream-skimming.

What this case establishes travels in direction even where its magnitudes do not. Contracting an entire publicly operated delivery network to private providers under capitation should be expected to contract per-capita inputs to the local healthcare sector; read in reverse, moving a system like Medicaid managed care toward direct government delivery would expand that sector without an assured gain in measured outcomes. The comparison here runs in favor of the private model on resource use, because Puerto Rico's public system carried more inputs per resident without delivering better measured outcomes; whether welfare improves depends on the pre-reform system's resourcing, the local provider-mobility option, and the incidence of the contraction, and the shares attributable to ownership, capitation, and access cannot be separated within a single reform that moved the three jointly. Three limitations bound these inferences. The four-year window suffices for the labor-market and short-horizon health outcomes but not for long-horizon mortality, health-capital accumulation, or the post-2002 multi-insurer carve-out structure (Centers for Medicare and Medicaid Services, 2002, 2004). The supply-side magnitude is concentrated in San Juan and sensitive to the §936 conditioning (Section 7), so we read it as the metro-concentrated, conservatively bounded effect rather than a uniform contraction, while the first stage survives both; identification of the contraction rests on the joint pattern of the policy-tied first stage, the heterogeneity-as-mechanism evidence, and a negative manufacturing-conditional estimate.

9 Tables

Table 1: County Descriptive Statistics by Year

	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001
Panel A: QCEW — Healthcare Labor Market †												
Employment per 1k pop.	2.86	3.28	3.44	3.61	3.90	4.03	4.54	5.02	5.52	5.92	6.22	7.67
	(4.74)	(5.15)	(5.05)	(5.23)	(5.41)	(5.33)	(5.56)	(5.85)	(6.49)	(6.76)	(7.16)	(9.94)
Establishments per 1k pop.	0.62	0.66	0.73	0.73	0.75	0.82	0.89	0.93	0.99	1.07	1.11	1.13
	(0.54)	(0.54)	(0.57)	(0.58)	(0.57)	(0.60)	(0.62)	(0.64)	(0.65)	(0.67)	(0.68)	(0.69)
Wage bill per 1k pop. ($)	5,405	6,558	7,059	7,454	7,966	8,478	9,490	10,651	11,852	12,772	13,439	16,981
	(10,809)	(12,859)	(13,280)	(13,985)	(13,962)	(14,211)	(15,151)	(16,101)	(18,296)	(19,145)	(20,646)	(26,925)
Counties	59	59	59	59	59	59	59	59	59	59	59	59
Treated Counties	–	–	–	–	5	16	38	38	58	58	59	59
Panel B: CMS — Provider of Services ‡												
Total Providers	–	202	222	242	248	254	259	262	265	268	274	275
Government	–	42	42	48	48	48	44	44	44	45	45	43
Private	–	160	180	194	200	206	215	218	221	223	229	232
Providers per 1k pop.	–	0.08	0.09	0.09	0.09	0.10	0.10	0.10	0.10	0.10	0.10	0.10
	(–)	(0.05)	(0.06)	(0.06)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
Beds per 1k pop.	–	3.17	3.15	3.21	3.18	3.18	3.16	3.10	3.08	3.18	3.20	3.21
	(–)	(3.08)	(3.06)	(3.06)	(3.04)	(3.09)	(3.14)	(3.09)	(3.07)	(3.28)	(3.26)	(3.26)
Services per 1k pop.	–	0.46	0.48	0.48	0.49	0.49	0.49	0.48	0.47	0.48	0.48	0.49
	(–)	(0.47)	(0.49)	(0.49)	(0.49)	(0.50)	(0.49)	(0.47)	(0.47)	(0.47)	(0.47)	(0.51)
FTE per 1k pop.	–	9.04	9.17	9.54	12.85	12.54	12.41	12.33	12.17	12.10	12.36	12.61
	(–)	(8.34)	(8.44)	(8.45)	(19.54)	(19.46)	(18.40)	(18.09)	(17.96)	(18.14)	(18.96)	(19.14)
Counties	–	40	40	40	40	40	40	40	40	40	40	40
Treated Counties	–	–	–	–	5	10	29	29	39	39	40	40
Panel C: VS — Demographic Flow, Hospital Ownership, and Patient Health ¶												
Population	45,228	45,502	45,884	46,435	47,253	47,684	47,863	48,792	49,147	49,571	48,935	49,228
	(59,205)	(59,564)	(60,064)	(60,778)	(61,282)	(59,393)	(58,842)	(59,597)	(59,858)	(60,151)	(59,129)	(58,987)
Crude birth rate per 1k pop.	19.38	18.45	18.54	18.66	18.08	17.46	17.42	17.42	16.19	15.69	15.64	14.64
	(2.38)	(2.01)	(2.24)	(1.98)	(1.86)	(2.11)	(2.32)	(2.19)	(1.94)	(2.21)	(1.44)	(1.48)
Crude death rate per 1k pop.	7.02	6.97	7.19	7.43	7.28	7.67	7.57	7.44	7.55	7.23	7.16	7.21
	(1.14)	(1.01)	(1.01)	(1.06)	(1.20)	(1.31)	(1.28)	(1.19)	(1.15)	(1.10)	(1.11)	(1.00)
Crude net migration rate per 1k pop.	–	-5.51	-3.04	0.66	8.56	10.90	-2.87	13.48	-1.47	1.91	-9.89	1.04
	(–)	(2.28)	(2.72)	(2.51)	(29.54)	(67.48)	(26.57)	(20.11)	(17.10)	(17.97)	(78.74)	(8.35)
Government hospital birth share (%)	65.8	65.9	65.3	64.3	63.2	59.1	51.6	45.2	36.6	33.6	30.6	32.1
	(11.3)	(11.2)	(10.9)	(10.7)	(10.7)	(12.1)	(15.2)	(15.3)	(14.6)	(18.6)	(19.8)	(22.7)
Private hospital birth share (%)	33.8	33.8	34.4	35.5	36.5	39.6	48.1	54.6	63.1	66.2	69.2	67.8
	(11.4)	(11.3)	(11.0)	(10.7)	(10.9)	(11.6)	(15.3)	(15.3)	(14.6)	(18.6)	(19.9)	(22.8)
Infant mortality rate per 1k live births	12.74	12.67	12.71	13.40	11.84	14.28	9.91	12.10	10.28	10.79	10.15	9.62
	(6.09)	(5.30)	(4.47)	(5.45)	(4.97)	(7.33)	(4.90)	(6.42)	(5.13)	(4.30)	(5.45)	(5.04)
Low birth weight rate (%)	9.28	9.25	9.48	9.64	9.85	10.12	10.49	11.06	10.98	11.36	10.96	11.48
	(1.62)	(1.79)	(2.02)	(1.66)	(1.43)	(1.90)	(1.90)	(2.32)	(2.24)	(2.46)	(2.48)	(4.07)
Counties	78	78	78	78	78	78	78	78	78	78	78	78
Treated Counties	–	–	–	–	10	22	52	52	77	77	78	78
Panel D: PR DOL — Macro Labor Market #												
Labor force / pop. (%; Figure 3)	30.5	31.4	32.0	32.2	31.4	31.8	32.4	32.5	32.3	31.3	31.5	30.9
	(4.6)	(4.7)	(5.1)	(5.1)	(4.8)	(4.9)	(5.1)	(5.0)	(5.2)	(5.2)	(4.4)	(4.3)
Unemployment rate (%; U / LF)	16.9	18.4	19.3	20.1	17.1	16.0	15.8	16.1	15.8	14.2	11.6	13.1
	(4.7)	(4.9)	(5.1)	(5.1)	(4.6)	(4.7)	(4.5)	(4.8)	(4.6)	(4.7)	(2.2)	(2.7)
Counties	78	78	78	78	78	78	78	78	78	78	78	78
Treated Counties	–	–	–	–	10	22	52	52	77	77	78	78

Means with standard deviations in parentheses. Sample is the PR balanced-panel counties (78 total); per-dataset counts below). † *Panel A (QCEW):* Overall Healthcare = NAICS 621 (Ambulatory) + 622 (Hospitals) + 623 (Nursing/Residential). All wage variables are CPI-deflated using 1982-1984 as the base year. ‡ *Panel B (CMS):* Provider counts are point-in-time totals from the CMS Provider of Services file; Government / Private split is the contemporaneous ownership classification. The trajectory rows (Always Government, Always Private, Gov. → Private switchers) classify each provider over its full panel history and are stable across years. CMS balanced panel 1991–2000: 40 counties.*CMS provider mix (provider-years, 1991–2000):* The CMS county panel summarizes 2,586 provider-year observations from 296 unique facilities. By ownership: 454 (17.6%) government, 872 (33.7%) private for-profit, 1,024 (39.6%) private nonprofit, 236 (9.1%) private-unspecified. By facility category, 36.8% are Hospitals (CMS POS code 01); the remaining provider categories are Hospital (36.8%); Home Health Agency (19.8%); Hospice (18.1%); ESRD Facility (11.9%); Ambulatory Surgical Center (7.8%); Other (5.6%). ¶ *Panel C (VS):* Crude birth rate is live births per 1,000 mid-year population. Crude death rate is deaths per 1,000 mid-year population. Crude net migration rate is the residual of the demographic-balancing equation: $[(\text{Pop}_t - \text{Pop}_{t-1}) - (\text{Births}_t - \text{Deaths}_t)]/\text{Pop}_{t-1} \times 1000$, NA in 1990 (no $t-1$ population). Government and private hospital birth shares are percentages of all live births. Low birth weight rate is the percentage of live births under 2,500g. VS balanced panel 1990–2001: 78 counties. [2pt]# *Panel D (PR DOL):* Labor force / pop. is labor force divided by total county population × 100; employment / pop. is employment divided by total county population × 100. Both use the non-standard denominator (total population) rather than the BLS-standard civilian non-institutional population aged 16+ (CNIWAP), which is not available at the county level for the 1990–2001 panel. The unemployment rate is reported as BLS-standard (unemployed / labor force × 100). PR DOL balanced panel 1990–2001: 78 counties. Across all 936 county-year observations: BLS-style U/LF mean = 16.19%; alternative U/Pop mean = 5.04%; difference = 11.16 pp; Pearson correlation between the two series = 0.878. The two series differ in level but not in trends. *Pre-reform (1990–1993) cross-county medians used as thresholds in heterogeneity figures (Main Figure 5; Appendix Figures A.5–A.7; Online Appendix Figures OA.21–OA.22):* Government hospital birth share = 63.2%; Private hospital birth share = 36.3%; Healthcare employment per 1k pop. = 1.51; Population = 30,362; Unemployment rate = 18.6%. Each median is computed across counties after first averaging the outcome over the 1990–1993 window within each county.

Table 2: Pre-Reform Balance Across Treatment Cohorts (1990–1993 Means)

Variable	Cohort 1 (1994) *Mean*	Cohort 2 (1995) *Mean*	Cohort 3 (1996) *Mean*	Cohort 4 (1998) *Mean*	Cohort 5 (2000) *Mean*	F-test *p-value*	N
Total population	43,889	34,467	37,871	45,489	443,627	<0.001	78
Log population	10.124	10.287	10.330	10.427	13.003	0.013	78
Share urban (top-20 by 1990 density)	0.600	0.417	0.433	0.560	1.000	0.638	78
Gov. hospital birth share (%)	73.57	66.70	67.17	59.30	61.64	0.003	78
HC employment per 1k pop.	4.01	2.28	2.64	2.87	29.72	<0.001	60
Labor force participation rate (%)	34.96	29.79	28.96	33.90	35.47	<0.001	78
Unemployment rate (%)	15.23	19.60	20.97	17.12	10.70	<0.001	78
Crude net migration rate (per 1k)	-3.80	-2.87	-3.25	-1.48	1.89	0.002	78
Infant mortality rate (per 1k)	13.59	11.21	13.74	12.10	13.02	0.139	78
Crude birth rate (per 1k)	19.90	19.11	19.39	17.43	17.22	<0.001	78
Agriculture, forestry & fishing emp. per 1k	1.451	16.195	8.422	15.991	–	0.393	25
Transportation & warehousing emp. per 1k	3.595	0.623	0.644	1.287	9.811	0.037	78
Arts, entertainment & recreation emp. per 1k	4.745	–	0.281	0.342	2.145	0.525	11
Counties	10	12	30	25	1		78

Cell entries are 1990–1993 means of pre-reform covariates by treatment cohort. Cohorts 1–5 correspond to reform-rollout years 1994, 1995, 1996, 1998, and 2000 respectively (no rollouts in 1997 or 1999). The F-test column reports the p-value of the one-way ANOVA test for the joint null hypothesis that all five cohort means are equal (`aov(var ~ factor(cohort))`); a small p-value indicates that the listed covariate differs systematically across cohorts and is therefore not balanced under the rollout schedule. **Sources.** Population, gov. hospital birth share, infant mortality, crude birth rate, and crude net migration rate are 1990–1993 means from the Puerto Rico Department of Health vital statistics. Healthcare employment per 1,000 residents (NAICS Overall Healthcare = 621 + 622 + 623) and the three placebo-industry rows (Agriculture/Forestry/Fishing = NAICS 11; Transportation & Warehousing = NAICS 48-49; Arts/Entertainment/Recreation = NAICS 71) are 1990–1993 means computed from the QCEW (BLS) county-quarter panel: aggregated across ownership classes, averaged to annual frequency requiring at least 3 reporting quarters per year, then divided by the population. Labor force participation and unemployment rates are 1990–1993 means from the PR Department of Labor. The urban indicator equals one for the 20 counties with the highest 1990 population density and zero otherwise; the cohort-cell entry is the share of urban counties within the cohort. The three placebo-industry rows (NAICS 11, 48-49, 71) are the same industries used in the QCEW placebo robustness analysis (Online Appendix Section 7)

Table 3: Effects of the Reform on Hospital Ownership Composition (First Stage)

	(1) Gov. Hospital Birth Share (%)	(2) Priv. Hospital Birth Share (%)
ATT	-14.79***	15.19***
	(1.94)	(1.93)
Pre-treatment mean of dep. var.	62.55	37.04
ATT as % of pre-treatment mean	-23.6%	41.0%
N counties	78	78

Estimates use the Callaway and Sant'Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\, p < 0.10$, $^{**}\, p < 0.05$, $^{***}\, p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). Government and private hospital birth shares are defined as the count of live births occurring at facilities of each ownership class divided by the total count of live births in the county-year, expressed in percent. The two shares are not exact complements because a small tail of births (typically <1%) is recorded as non-hospital or with unspecified facility ownership.

Table 4: Reform Effects on Healthcare Employment, Wages, and Establishments

	Healthcare (NAICS 621+622+623)			NAICS 621 (Ambulatory)		
	(1) Empl. per 1k	(2) Wages/Empl. per 1k	(3) Estabs. per 1k	(4) Empl. per 1k	(5) Wages/Empl. per 1k	(6) Estabs. per 1k
ATT	-0.511***	0.630	-0.050***	-0.502***	0.867	-0.034*
	(0.157)	(3.734)	(0.019)	(0.125)	(3.682)	(0.018)
Pre-treatment mean of dep. var.	3.923	38.962	0.759	2.895	38.889	0.747
ATT as % of pre-treatment mean	-13.0%	1.6%	-6.6%	-17.3%	2.2%	-4.6%
N counties	59	59	59	59	59	59

Estimates use the Callaway and Sant'Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\, p < 0.10$, $^{**}\, p < 0.05$, $^{***}\, p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). The reference period for event-time aggregation is $t = -1$ (year before treatment).

Table 5: Effects of The Reform on Patient Health (Short-Horizon) and Hospital Capacity

	Patient Health (VS)		Hospital Capacity (CMS)	
	(1) IMR (per 1k live births)	(2) Low Birth Weight (%)	(3) Beds per 1k pop.	(4) Services per 1k pop.
ATT	-1.205	0.060	-0.265***	-0.024**
	(0.928)	(0.230)	(0.065)	(0.010)
Pre-treatment mean of dep. var.	12.620	9.717	3.389	0.487
ATT as % of pre-treatment mean	-9.5%	0.6%	-7.8%	-4.9%
N counties	78	78	40	40

Estimates use the Callaway and Sant'Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). The reference period for event-time aggregation is $t = -1$ (year before treatment).

Table 6: Worker-Side Adjustment Margins to The Reform

	(1) Labor force / pop. (%)	(2) Crude Net Migration (per 1k)	(3) Crude Birth Rate (per 1k)
ATT	-1.110***	-14.651**	-0.564**
	(0.241)	(7.389)	(0.225)
Pre-treatment mean of dep. var.	31.895	1.059	18.267
ATT as % of pre-treatment mean	-3.5%	-1382.9%	-3.1%
N counties	78	78	78

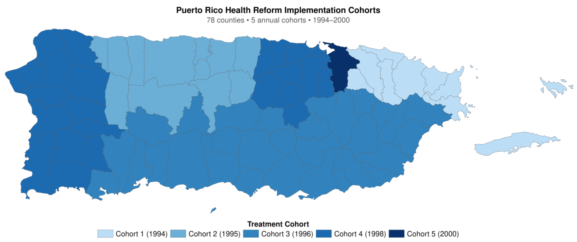
Estimates use the Callaway and Sant'Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). The reference period for event-time aggregation is $t = -1$ (year before treatment).

Table 7: Heterogeneity in the Effects by Pre-Reform Public-Hospital Exposure

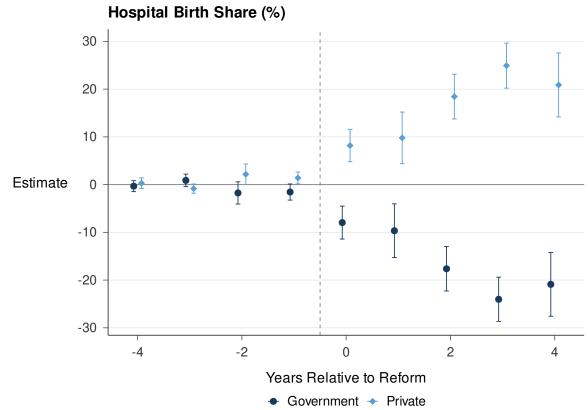
	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
Below-median pre-reform government hospital birth share				
ATT	-15.49***	-0.346	-9.206	-1.454
	(2.44)	(0.233)	(10.814)	(1.101)
Pre-treatment mean	55.49	4.911	2.772	11.577
ATT as % of pre-mean	-27.9%	-7.0%	-332.1%	-12.6%
N counties	39	36	39	39
Above-median pre-reform government hospital birth share				
ATT	2.41	0.145	-23.891**	0.660
	(9.15)	(0.268)	(12.047)	(1.329)
Pre-treatment mean	71.21	2.069	-1.126	13.925
ATT as % of pre-mean	3.4%	7.0%	2121.2%	4.7%
N counties	39	23	39	39

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, based on two-sided tests with cluster-robust standard errors. counties are partitioned at the cross-county median of the 1990–1993 pre-reform government hospital birth share (median = 63.24%). *Below-median* (39 counties) = lower pre-reform reliance on the public hospital system; *Above-median* (39 counties) = higher pre-reform reliance.

10 Figures

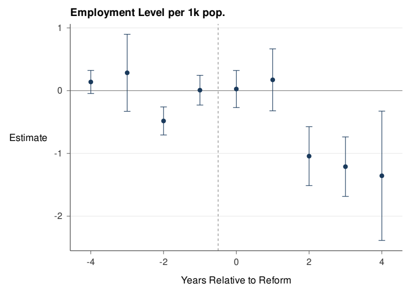


(a) Cohorts

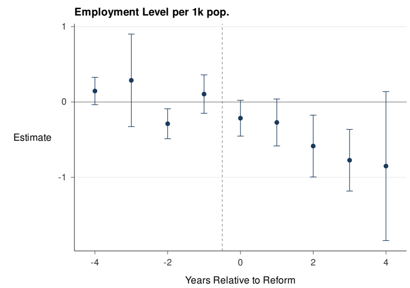


(b) Birth Share (pct)

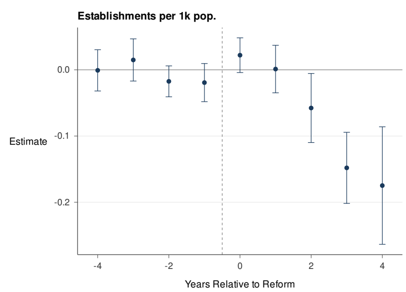
Figure 1: Policy Rollout



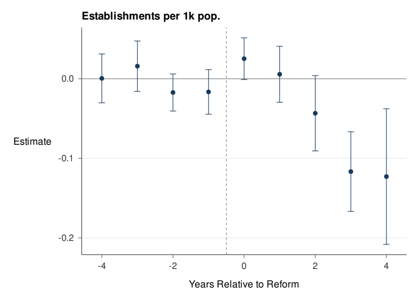
(a) Employment (per 1k) - 621+622+623



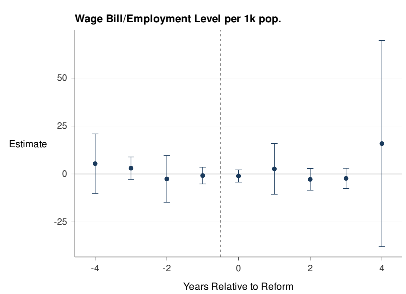
(b) Employment (per 1k) - 621



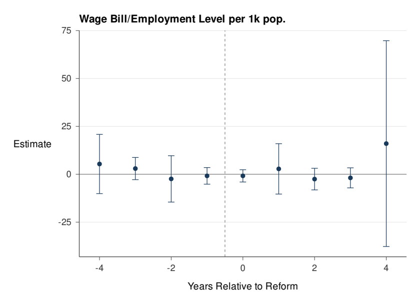
(c) Establishments (per 1k) - 621+622+623



(d) Establishments (per 1k) - 621

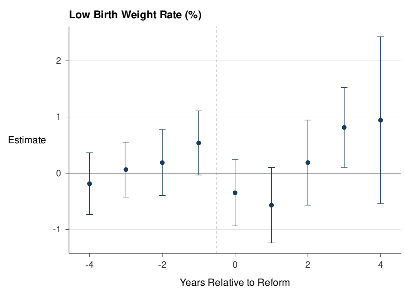


(c) Wage Bill/Employee (per 1k) - 621+622+623

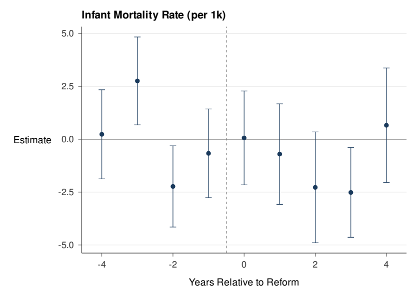


(d) Wage Bill per Employee (per 1k) - 621

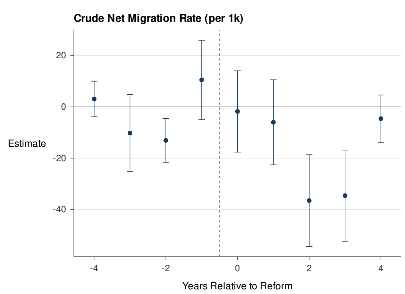
Figure 2: Healthcare Labor-Market Contraction



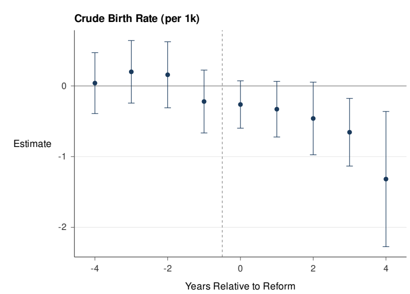
(a) Low Birth Weight



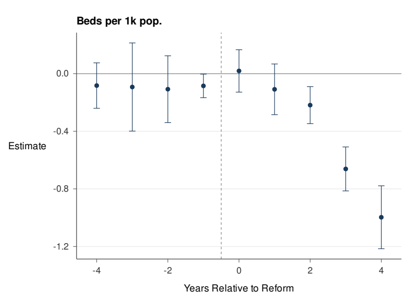
(b) Infant Mortality Rate



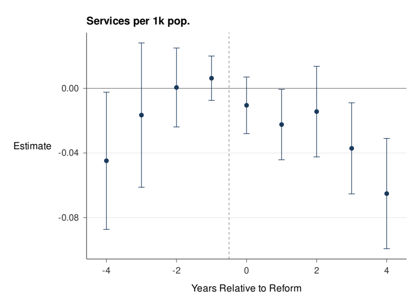
(c) Net Migration Rate



(d) Birth Rate

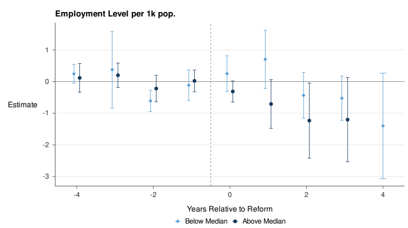


(e) Beds

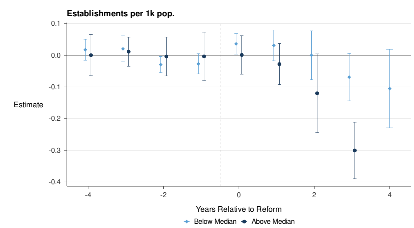


(f) Services

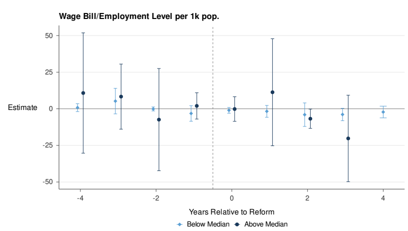
Figure 3: Patient Health



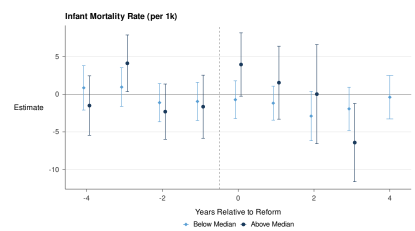
(a) Employment (per 1k) - 621+622+623



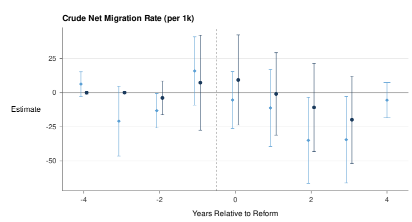
(b) Establishments (per 1k) - 621+622+623



(c) Wage Bill per Employee (per 1k) - 621+622+623

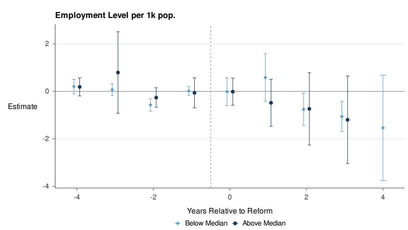


(d) Infant Mortality Rate (per 1k)

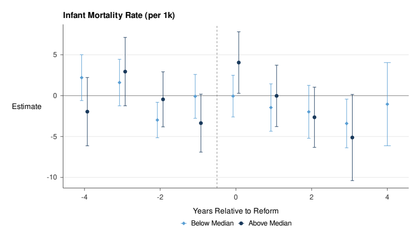


(d) Net Migration Rate

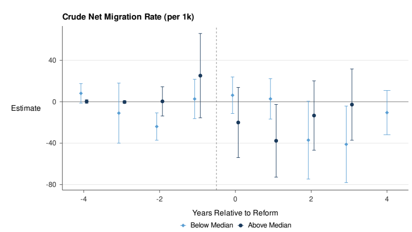
Figure 4: Heterogenous Effects by Poverty Rate



(a) Employment (per 1k) - 621+622+623



(b) Infant Mortality Rate (per 1k)



(c) Net Migration Rate (per 1k)

Figure 5: Heterogeneous Effects by Pre-Reform Gov. Hospital Exposure

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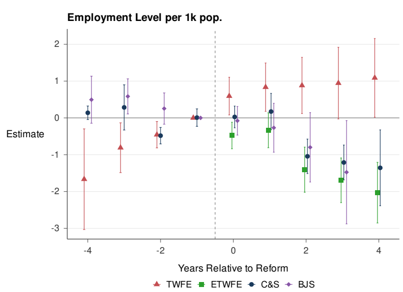
Appendix

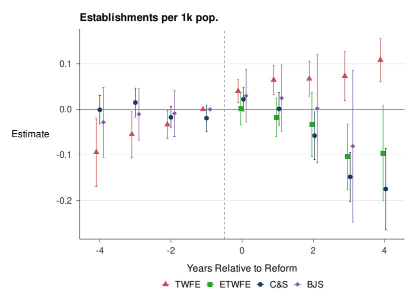
The Appendix collects the supplementary tables referenced in the body. Each table is referenced from a footnote to the relevant subsection of the main text. The Online Appendix (Section 10) hosts the remaining robustness, sample-restriction, and subgroup specifications.

A.1 Estimator-Class Robustness

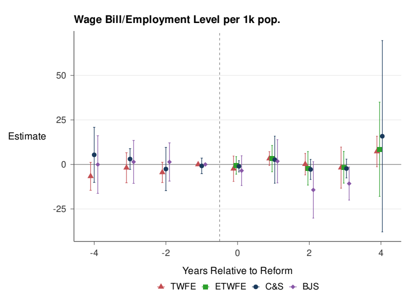
This subsection reports a four-estimator triangulation on the six primary outcomes (TWFE; the ETWFE estimator of Wooldridge, 2025; the doubly robust estimator of Callaway & Sant'Anna, 2021, C&S; and Borusyak et al., 2024, BJS). Pre-treatment parallel trends hold for the labor-market outcomes under all four estimators. The ETWFE, C&S, and BJS estimators agree on the sign and magnitude of the first-stage hospital-ownership share, healthcare employment, labor-force participation, and the hospital-bed contraction. TWFE reverses sign on the outcomes with the most treatment-effect heterogeneity (healthcare employment, infant mortality, and the low-birth-weight share). The crude net migration rate is the only outcome on which the four estimators disagree on sign.



(a) Employment Level per 1k



(b) Establishments per 1k



(c) Wage Bill per Employment per 1k

Figure A.1: Overall Healthcare Labor Market - Estimator Comparison

Table A.1: Four-Estimator Robustness: ATT Estimates Under TWFE, ETWFE, C&S, and BJS

	(1) TWFE	(2) ETWFE	(3) C&S	(4) BJS
Gov. Hospital Birth Share (%)				
ATT	-8.08***	-16.52***	-14.79***	-11.25***
	(1.89)	(1.60)	(1.94)	(1.69)
ATT as % of pre-mean	-12.9%	-26.4%	-23.6%	-18.0%
Pre-treatment mean			62.55	
counties			78	
HC Employment per 1k pop. (NAICS 621+622+623)				
ATT	0.868**	-1.008***	-0.511***	-0.655***
	(0.352)	(0.181)	(0.157)	(0.217)
ATT as % of pre-mean	22.1%	-25.7%	-13.0%	-16.7%
Pre-treatment mean			3.923	
counties			59	
Labor force / pop. (%)				
ATT	0.059	-1.155***	-1.110***	-0.712***
	(0.350)	(0.231)	(0.241)	(0.235)
ATT as % of pre-mean	0.2%	-3.6%	-3.5%	-2.2%
Pre-treatment mean			31.895	
counties			78	
Crude Net Migration Rate (per 1k pop.)				
ATT	-5.689	-1.525	-14.651**	4.513
	(15.281)	(2.931)	(7.389)	(3.523)
ATT as % of pre-mean	-537.1%	-144.0%	-1382.9%	426.0%
Pre-treatment mean			1.059	
counties			78	
Infant Mortality Rate (per 1k live births)				
ATT	1.973**	-1.524**	-1.205	-0.513
	(0.815)	(0.599)	(0.928)	(0.727)
ATT as % of pre-mean	15.6%	-12.1%	-9.5%	-4.1%
Pre-treatment mean			12.620	
counties			78	
Beds per 1k pop.				
ATT	-0.387	-0.443***	-0.265***	-0.189***
	(0.333)	(0.072)	(0.065)	(0.056)
ATT as % of pre-mean	-11.4%	-13.1%	-7.8%	-5.6%
Pre-treatment mean			3.389	
counties			40	

Notes: Four estimators are reported. TWFE: two-way fixed-effects regression with county and year FEs and event-time dummies. ETWFE: Wooldridge (2021) Extended Two-Way Fixed Effects. C&S: Callaway and Sant'Anna (2021). BJS: Borusyak, Jaravel & Spiess (2024). Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors.

A.2 Excluding San Juan

Reports the main-vs.-excluded-San-Juan specifications for the QCEW labor-market and CMS hospital-capacity outcomes. The healthcare-employment and bed-capacity estimates attenuate substantially when San Juan is removed; the first-stage ownership-transition estimate (Online Appendix Table OA.3) survives the exclusion at conventional significance.

Table A.2: Sample-Specification Robustness: Excluding San Juan and the Unbalanced-Panel Variant

	HC Empl. per 1k pop.			Beds per 1k pop.			Infant Mortality Rate		
	(1) Bal.	(2) Excl. SJ	(3) Unbal.	(4) Bal.	(5) Excl. SJ	(6) Unbal.	(7) Bal.	(8) Excl. SJ	(9) Unbal.
ATT	-0.511***	-0.168	-0.511***	-0.265***	0.090	-0.265***	-1.205	0.781	-1.205
	(0.157)	(0.191)	(0.157)	(0.065)	(0.140)	(0.065)	(0.928)	(1.237)	(0.928)
Pre-treatment mean of dep. var.	3.923	3.161	3.892	3.389	3.033	2.980	12.620	12.624	12.620
ATT as % of pre-treatment mean	-13.0%	-5.3%	-13.1%	-7.8%	3.0%	-8.9%	-9.5%	6.2%	-9.5%
counties	59	58	60	40	39	54	78	77	78

Notes: Estimates use the Callaway and Sant'Anna (2021). Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, based on two-sided tests with cluster-robust standard errors.

A.3 Section 936 Manufacturing Control

Reports the main healthcare-employment estimate with and without conditioning on pre-reform manufacturing exposure (1990–1993 mean of NAICS 31–33 employment per thousand at the county level). The C&S no-control estimate on the 54-county sample is -14.7 percent of pre-mean ($p < 0.01$); the C&S estimate with the strictly-pre-1996 manufacturing-exposure covariate is -6.8 percent of pre-mean (statistically insignificant). The TWFE columns are reported for direct horse-race comparison and return the positive point estimates expected under Goodman-Bacon contamination on heterogeneous-effect outcomes. The main contraction survives conditioning on §936 exposure at attenuated magnitude.

Table A.3: IRS Section 936 Manufacturing-Control Robustness: Healthcare Employment per 1,000 Residents under TWFE and Callaway-Sant'Anna, with and without Manufacturing Exposure as a Control

	TWFE		Callaway-Sant'Anna	
	(1) No control	(2) With manuf. control	(3) No control	(4) With baseline manuf. exposure
ATT on HC empl. per 1k	0.173	0.182	-0.484***	-0.224
	(0.198)	(0.215)	(0.164)	(0.233)
ATT as % of pre-mean	5.3%	5.5%	-14.7%	-6.8%
Coef. on contemp. manuf. per 1k	—	-0.003	—	—
	—	(0.013)	—	—
counties	54	54	54	54

Notes: Cluster-robust standard errors at the county level. Outcome: healthcare employment per 1,000 residents (NAICS 621+622+623). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. **Estimators.** Columns 1–2 report TWFE event-study estimates with county and year fixed effects.

A.4 Welfare Bounds

Translates the main supply contraction and the patient-health response into comparable units (1982–1984 USD, with parenthetical translations into 2025 USD). The cells span two parameter dimensions: the healthcare-employment-contraction estimate (main C&S versus §936-conditional C&S, the latter from Table A.3) and whether the patient-health response is valued at zero (the full-panel IMR null) or at the high-substitution-half magnitude. Because the full-panel infant-mortality effect is insignificant, the health-valued cells enter only as an upper bound and are evaluated across the four high-substitution partitions of Section 6.3, whose above-median IMR estimates range from -2.03 to -3.12 deaths per thousand live births. The zero-health cells, which rest entirely on the avoided supply cost, are \$300.8 million under the headline supply estimate (\$967 million in 2025 dollars; \$21.5 per resident per year) and \$262.5 million under the §936-conditional estimate (\$844 million in 2025 dollars; \$18.7 per resident per year). The health-valued cells add the monetized mortality reduction and, across the four partitions, range from \$1.05 to \$1.46 billion in 1982–1984 dollars (\$3.4 to \$4.7 billion in 2025 dollars). The exercise is partial-equilibrium and back-of-envelope, not a structural welfare calculation.

Table A.4: Welfare Bounds: Cost Savings, Patient-Health Benefits, and Net Welfare

	Lower bound (ΔIMR = 0) *(1982-84 \$M)*	Upper bound (ΔIMR high-substitution half) *(1982-84 \$M)*
Panel A: Headline ATT (Tables 2 and 4)		
Cost saving (\$M)	300.8	300.8
Health benefit (\$M)	0.0	958.7
Net welfare (\$M)	**300.8**	**1,259.5**
Per capita per year (\$)	21.5	90.0
Panel B: § 936-Conditional ATT (Table A.4 col. 4)		
Cost saving (\$M)	262.5	262.5
Health benefit (\$M)	0.0	544.1
Net welfare (\$M)	**262.5**	**1,221.2**
Per capita per year (\$)	18.7	87.2

Notes: Partial-equilibrium back-of-envelope welfare bound; **not** a structural calculation. Body cells are in real 1982-84 USD (CPI-U all-items, 1982-84 = 100); the headline upper bound corresponds to approximately \$4,048 M in 2025 USD. **Cost-saving inputs.** Δbeds/1k = -0.265 (Table 4); ΔHC empl./1k = -0.511 headline (Table 2) and -0.224 § 936-conditional (Table A.4 col. 4). Cost per bed-year (\$62,756, 1982-84 USD) is the implied 1990–1993 mean hospital wage bill from CMS POS and QCEW NAICS 621+622+623, scaled by 2.1 (U.S. HCRIS hospital total-cost-to-wage-bill ratio). **Patient-health benefit.** VSL = \$10 M (U.S. EPA standard, 2024 USD), rebased to 1982-84 USD as $\approx$ \$3.19 M. ΔIMR = -1.47 per 1,000 live births in the high-substitution half (mean across pre-reform population density, total population, HC employment intensity, and commuting inflow; Tables A.7, OA.9–OA.11); a 0.5 multiplier rescales island-wide births to the high-substitution half.

A.5 Spillover-Excluded Sample (QCEW)

Reports the QCEW labor-market specifications on the non-border subsample ($n = 27$ counties retained out of 78). The healthcare-employment, wage, and establishment estimates are statistically indistinguishable from zero on this restricted sample; the first-stage hospital-ownership-share estimate (Online Appendix Table OA.5) survives the restriction at -12.14 percentage points ($p < 0.01$, -19.4 percent of pre-mean).

Table A.5: Spillover-Excluded Sample: QCEW Healthcare Headline Outcomes

	(1) Empl. per 1k	(2) Wages/Empl. per 1k	(3) Estabs. per 1k
ATT	-0.054	3.083	0.021
	(0.318)	(9.307)	(0.037)
Pre-treatment mean of dep. var.	3.193	42.066	0.787
ATT as % of pre-treatment mean	-1.7%	7.3%	2.7%
counties	27	27	27

Notes: Callaway and Sant'Anna (2021) doubly-robust ATT with not-yet-treated controls; reference period $t = -1$; cluster-robust standard errors at the county level via multiplier bootstrap (1,000 reps). Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$. **Spillover-exclusion sample.** Restricted to the 27 counties that share no boundary with a different-cohort county.

A.6 Heterogeneity by Pre-Reform Population Density

Partitions the seventy-eight counties at the median pre-reform population density. The infant-mortality estimate in the above-median (urban) cell is -2.027 per thousand live births ($p < 0.05$, -16.9 percent of pre-mean). The first of four heterogeneity dimensions on which the IMR signal is stronger than the baseline-exposure cut reported in Table 7.

Table A.6: Heterogeneity by Pre-Reform Population Density (Urban vs. Rural Median Split)

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
Below-median pre-reform population density (urban vs. rural)				
ATT	-1.27	-0.105	-7.565	2.436
	(5.27)	(0.262)	(10.268)	(1.956)
Pre-treatment mean	67.13	2.074	0.587	13.273
ATT as % of pre-mean	-1.9%	-5.1%	-1288.3%	18.4%
counties	39	24	39	39
Above-median pre-reform population density (urban vs. rural)				
ATT	-18.21***	-0.418*	-12.513	-2.027**
	(2.36)	(0.219)	(11.042)	(0.892)
Pre-treatment mean	58.11	5.172	1.514	11.993
ATT as % of pre-mean	-31.3%	-8.1%	-826.8%	-16.9%
counties	39	35	39	39

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). Pre-reform population density = (1990–1993 mean municipio population) / (land area in km^2). Counties partitioned at the cross-municipio median; below-median ('Rural') = 24 counties, above-median ('Urban') = 35 counties.

A.7 Per-Capita versus Raw-Level Specifications

Reports per-thousand-resident and $\log(1+y)$ raw-level specifications side-by-side for QCEW healthcare employment, establishments, and the wage bill. Healthcare employment per thousand falls by 13.0 percent ($p < 0.01$); $\log(1+y)$ raw-level employment grows by approximately 19.0 percent ($p < 0.05$). The per-capita decline reflects population growth that outpaced the growth of the local healthcare workforce; the per-capita normalization is the welfare-relevant scaling adopted throughout (Section 3). The distributional finding of the paper rests on the per-capita denominator and is not a feature of the absolute headcount.

Table A.7: QCEW Healthcare Headline Outcomes: Per-1,000 vs. Log-Level

	HC Employment		HC Establishments		HC Wage Bill	
	Per 1,000	Log(1+x)	Per 1,000	Log(1+x)	Per 1,000	Log(1+x)
ATT	-0.511***	0.190**	-0.050***	0.144***	0.630	0.247
	(0.157)	(0.074)	(0.019)	(0.048)	(3.734)	(0.263)
Pre-treatment mean	3.923	4.293	0.759	3.161	38.962	11.000
Implied % effect	-13.0%	19.02%	-6.6%	14.38%	1.6%	24.69%

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors.

A.8 Labor Force

The labor-force participation rate contracts over the rollout period as the reform takes effect, consistent with the negative labor-force-participation estimate in Section 6.2. Because the unemployment rate does not move over the same window, the adjustment operates on participation rather than on measured unemployment.

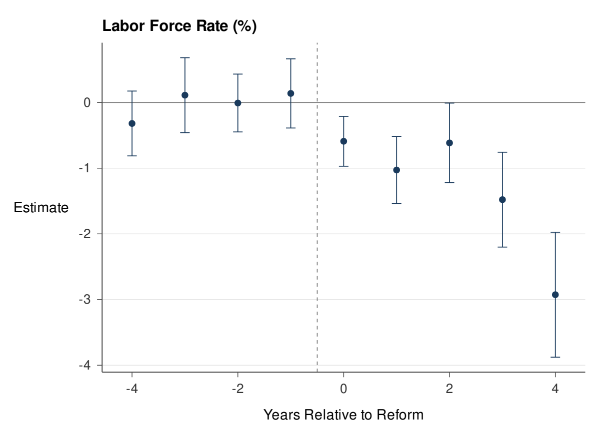


Figure A.2: Labor Force

A.9 Financing Detail

The 1993 reform replaced the Department of Health line-item appropriation that had funded the pre-reform delivery system with a multi-source aggregate funding pool flowing through ASES to the contracting insurers. Realized program financing for fiscal year 2002–2003 totaled approximately $1.282 billion, drawn in approximate proportions of seventy-three percent from the Commonwealth General Fund, fifteen percent from federal Medicaid, eleven percent from the counties, and two percent from the State Children's Health Insurance Program (SCHIP) (Pan American Health Organization, 2007). The federal Medicaid share is constrained by the Section 1108(g) territorial Medicaid cap and the fixed Federal Medical Assistance Percentage (FMAP) applicable to Puerto Rico, both of which produce a federal contribution per enrollee substantially below what Puerto Rico would receive under the income-conditioned FMAP formula applied to the fifty states (Kaiser Family Foundation, 2024b; Medicaid and CHIP Payment and Access Commission, 2020). The program continues to operate under the names *Mi Salud* and *Vital*, covering approximately 1.3 million Medicaid and CHIP enrollees as of 2026 (Centers for Medicare and Medicaid Services, 2026).

A.10 Data Construction

This subsection reports the construction details for each of the six data sources summarized in Section 3.

QCEW (1990–2000). The Quarterly Census of Employment and Wages provides county-quarter establishment cells with average monthly employment, total quarterly wages, and active establishment counts by NAICS code and ownership class (federal, state, and local government, and private). The annualization procedure proceeds in two steps. Within each county-quarter cell, employment, wages, and establishments are summed across ownership classes to produce a quarter-level total. Across quarters within a county-year, employment and establishments are averaged and wages are summed; cells with fewer than three reporting quarters are dropped on the grounds that they reflect missing rather than zero healthcare activity. Per-thousand-resident outcomes are constructed by dividing each headcount by the contemporaneous Vital Statistics population in thousands. The wage-per-employee intensive-margin outcome is the annualized real wage bill divided by annualized employment, expressed in 2024 dollars using BLS CPI-U annual-average deflators. The balanced-cohort QCEW healthcare panel covers 59 counties and 649 county-year observations.

CMS Provider of Services (1991–2000). The CMS POS public-use files provide annual provider-level records of Medicare-certified facilities with full-time-equivalent counts, certified beds, services rendered, and ownership classification (government, voluntary nonprofit, proprietary). Provider-level records are aggregated to the county-year level by summing across providers operating in each county. Of 296 unique providers tracked across the 1991–2000 panel, 32 are continuously government-owned, 225 are continuously private, and 12 switched ownership within the panel window; the small switcher count underscores the partial coverage of the privatization shock at the Medicare-certified layer. The balanced CMS Track A panel covers 40 counties and 400 county-year observations.

Vital Statistics (1990–2000). The Puerto Rico Department of Health Vital Statistics annual records provide county-year totals of live births, deaths, infant deaths, and end-of-year population. These records support construction of (i) the crude birth rate, (ii) the infant mortality rate, (iii) the low-birth-weight share, (iv) the government and private hospital birth shares, (v) the crude net migration rate (CNMR), constructed as the demographic-balancing residual

$$\text{CNMR}_{i,t} \;=\; \frac{(\text{Pop}_{i,t} - \text{Pop}_{i,t-1}) - (\text{Births}_{i,t} - \text{Deaths}_{i,t})}{\text{Pop}_{i,t-1}} \times 1{,}000, \tag{4}$$

and (vi) total county population in levels and in logarithms, used as a complementary demographic outcome to (4) that is not constructed as a residual and is therefore less sensitive to compounding measurement error. CNMR estimates are reported in level units (per thousand residents per year) rather than as a percentage of the pre-treatment mean because the CNMR pre-treatment mean is close to zero (+1.06 per thousand). The CNMR is reported alongside total population because the two outcomes can in principle disagree on the sign of the local demographic response if the constituents (births, deaths, populations) are measured with component-wise error; the joint reading is informative about which margin (residual versus level) is the more reliable identification anchor in any given specification.

PR Department of Labor (1990–2000). The Puerto Rico Department of Labor and Human Resources monthly county-level labor-force survey (*Encuesta de Vivienda*) provides annualized county-year measures of the labor force, employment, and unemployment. The unemployment rate is constructed in BLS-standard form (unemployed over labor force). The labor-force-participation rate and the employment-to- population rate are constructed against total county population, since the BLS-standard 16-and-over civilian non-institutional denominator is not available at the Puerto Rican county level for the 1990–2000 panel. The maintained assumption is that the ratio of working-age civilian non-institutional population to total population is not differentially affected by treatment within the analysis window; the partial effect identified on the total-population-denominator rate is then proportional to the partial effect that would be identified on the BLS-standard rate.

NVSS Natality (conception years 1993–2000). The National Center for Health Statistics National Vital Statistics System natality public-use files provide birth-level records with infant birth weight, gestational age at delivery, the count of prenatal visits, the trimester of prenatal-care initiation, the one- and five-minute Apgar scores, and maternal demographic and educational characteristics. Puerto Rico appears in the separate U.S. territory natality file, which begins with the 1994 birth cohort. The empirical specification assigns treatment by conception year, treating a birth if its conception year post-dates the treatment date in the county of maternal residence, rather than by delivery year, on the grounds that the relevant exposure of the prenatal-care production function is the conception-to-delivery window; indexing the 1994-onward birth cohorts this way yields the 1993–2000 conception window used here.

BRFSS (1996–2000). The Centers for Disease Control and Prevention Behavioral Risk Factor Surveillance System provides individual-level adult-survey microdata with self-reported insurance status, self-reported general health (a five-point ordinal scale), an indicator for whether the respondent had been unable to see a doctor in the past twelve months because of cost, and standard demographic covariates. The BRFSS Puerto Rico coverage begins in 1996, after Cohorts 1 and 2 had completed implementation; the BRFSS analysis is therefore restricted to Cohorts 3–5 (1996, 1998, and 2000), for which a pre-treatment observation window exists within the panel.

A.11 Framework Derivation

This subsection reports the formal derivations behind the three-component decomposition (1) stated in Section 4.

Ownership component. Following Hart et al. (1997), the noncontractible margin includes provider effort, time per patient, and continuity of care; the contractible margin is the per-enrollee budget that ASES pays each managed-care organization. Let $\theta \in [0, 1]$ index the relative weight of quality ($\theta = 1$) versus cost ($\theta = 0$) in the provider's objective; under Hart et al. (1997), $\theta_{\text{public}} > \theta_{\text{private}}$. Letting $L^*(\theta)$ denote optimal labor input, the model implies $L^{*\prime}(\theta) > 0$, so the ownership-component response on healthcare employment per capita satisfies

$$\Delta L^{\text{ownership}} \;=\; L^*(\theta_{\text{private}}) - L^*(\theta_{\text{public}}) \;\leq\; 0. \tag{5}$$

The same primitive admits the quality-deterioration corollary referenced in Section 4: noncontractible quality q satisfies $\Delta q \leq 0$ unless the public provider was itself underinvesting on the quality margin relative to the private optimum.

Capitation component. The pre-reform compensation arrangement paid providers fixed government salaries; the post-reform arrangement pays managed-care organizations prospective per-member-per-month capitation, with insurers in turn contracting bilaterally with provider networks. Let $C(L)$ denote the provider's expected cost as a function of network labor input. Under salary, $C'(L)$ is borne by the principal (the Department of Health); under capitation, it is borne by the managed-care organization. As Duggan (2004) documents for U.S. Medicaid HMOs, and as Aizer et al. (2007) and Marton et al. (2014) discuss, this shift in residual cost-bearing creates an incentive to economize on input use, so the capitation margin reinforces the ownership-margin prediction in (5): $\Delta L^{\text{capitation}} \leq 0$.

Access component. To the extent that the reform expanded effective insurance coverage among the medical-indigent population, it raised demand for insured care and, through the standard insurance-and-supply-stimulus mechanism (Arrow, 1963; Dillender, 2022; Finkelstein, 2007; Hackmann et al., 2025), predicts an expansion of the local healthcare labor market. The direction of the access-component effect on L^* depends on whether the binding margin is demand expansion ($\Delta L^{\text{access}} > 0$) or the closing of the public outside option, network confinement, and primary-care gatekeeping ($\Delta L^{\text{access}} \leq 0$); the sign is therefore an empirical question.

Mobility margin. The reform's labor-supply incidence is shaped by a feature distinctive to the Puerto Rican setting: the affected provider workforce holds U.S. citizenship and U.S.-credentialed professional licenses, so the legal cost of relocation to the mainland is zero. The framework therefore admits an additional adjustment margin closed off in the mainland-U.S. place-based literature (Autor, Dorn, & Hanson, 2013; Yagan, 2019): out-migration of the displaced workforce. The empirical analysis traces this margin through the demographic-balancing identity (4), jointly with the level of county population. The two demographic measures are reported together because each is informative about a distinct aspect of the local labor supply, and disagreement between them carries identification content rather than being averaged away.

A.12 Empirical Strategy: Technical Detail

This subsection reports the dynamic-aggregation formula and the TWFE comparator specification referenced in Section 5.

Dynamic event-study aggregation. The event-study figures aggregate the group-time treatment effects $\widehat{\text{ATT}}(g,t)$ from (2) into event-time coefficients $\widehat{\theta}_\ell$ by cohort-size weights:

$$\widehat{\theta}_\ell \;=\; \sum_{g\in\mathcal{G}} \frac{N_g}{\sum_{g'} N_{g'}}\,\widehat{\text{ATT}}(g,g+\ell), \qquad \ell \in \{-4,\ldots,+4\}, \tag{6}$$

where N_g is the number of counties in cohort g and the reference period is $\ell = -1$. Confidence bands are constructed via the multiplier bootstrap of Callaway and Sant'Anna (2021) clustered at the county level with 1,000 replications, applied separately at each event time.

TWFE comparator specification. The canonical two-way fixed-effects (TWFE) event-study reported as estimator-class robustness in Appendix 10 regresses $Y_{i,t}$ on county fixed effects α_i, year fixed effects λ_t, and event-time indicators relative to treatment:

$$Y_{i,t} \;=\; \alpha_i + \lambda_t + \sum_{\ell\neq -1} \beta_\ell\, \mathbf{1}\{t - G_i = \ell\} \;+\; \varepsilon_{i,t}, \tag{7}$$

with standard errors clustered at the county level and the reference period $\ell = -1$ omitted.

Doubly robust three-step procedure. The $\widehat{\text{ATT}}(g,t)$ estimator in (2) is the doubly robust difference-in-differences estimator of Callaway and Sant'Anna (2021), which combines an inverse-probability-weighted estimator on the not-yet-treated comparison group with an outcome-regression-based correction term. The estimator is consistent if either the propensity-score model or the outcome-regression model is correctly specified, providing a robustness margin over either component alone. Implementation uses the `did` package in R with the default propensity-score specification (logistic regression on a constant) and the default outcome-regression specification (county and year fixed effects); the group-time-by-event-time estimates are then aggregated via (6) for the event-study figures and via the post-treatment-cell-count-weighted average for the regression tables. The not-yet-treated control group is implemented as `control_group = "notyettreated"` in `did::att_gt`.

A.13 BRFSS Coverage by Subgroup

This subsection expands on the BRFSS access-component result reported in Section 6.3. The BRFSS 1996–2000 panel restricts the analysis to Cohorts 3–5, for which a pre-treatment observation window exists within the panel (Appendix 10). The single-cohort exposure pattern within the BRFSS window means that Callaway and Sant'Anna (2021) is not implementable on this sample; the BRFSS results are reported under the Wooldridge (2025) ETWFE estimator with TWFE comparator estimates also reported. Six demographic subgroups are analyzed (Online Appendix Tables OA.14–OA.19): ages 18–29, 30–44, 45–64, and 65 and older; college graduates; and employed respondents. Coverage rises across all six subgroups, with the largest point estimate of +10.5 percentage points in the 45–64 age group ($p < 0.01$) and the smallest of +2.8 percentage points among respondents aged 65 and older ($p < 0.01$, smaller in magnitude because near-universal Medicare coverage at this age leaves limited room for the reform to expand effective insurance). The six subgroup estimates jointly rule out the no-coverage-change null even with the limited cohort exposure in the BRFSS sample, and the cross-subgroup pattern is consistent with the reform expanding effective insurance among the working-age medical-indigent population most directly targeted by the reform's eligibility criteria.

Online Appendix

The Online Appendix collects the additional sample-restriction, estimator-comparison, and subgroup specifications referenced in the main text. The exhibits are organized by topic, in the order in which they are referenced in the body.

OA.1 Macro Labor-Market Outcomes

Reports the BLS-standard unemployment rate, the labor-force-participation rate, and the employment-to-population rate side by side. The unemployment rate is statistically indistinguishable from zero against a pre-reform mean of 17.770 percent; the LFP and employment-to-population rates both fall by 3.5 percent of pre-mean. The unemployment-rate null together with the LFP decline is the labor-supply-margin signature of Section 6.2.

OA.2 Log-Specification on QCEW Main

Parallel specification on $\log(1+y)$ per-1,000 normalizations and $\log(1+y)$ raw-level outcomes for the QCEW healthcare aggregate. The per-capita and raw-level estimates point in opposite directions because the local population grew faster than the local healthcare workforce during the analysis window; the main finding rests on the per-capita normalization.

Table OA.1: Macro Labor-Market Effects of the reform (Online Appendix)

	(1) Unemployment Rate (%)	(2) Employment Rate (%)	(3) Labor Force Rate (%)
ATT	-0.245	-0.926***	-1.110***
	(0.285)	(0.219)	(0.241)
Pre-treatment mean of dep. var.	17.770	26.331	31.895
ATT as % of pre-treatment mean	-1.4%	-3.5%	-3.5%
counties	78	78	78

Notes: Estimates use the Callaway and Sant'Anna (2021) ATT. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors.

Table OA.2: Log(1+x) Specification: QCEW Healthcare Headline Outcomes (Online Appendix)

	Per-1k Population (log1p)			Levels (log1p)		
	(1) Empl./1k	(2) Wages/Empl./1k	(3) Estabs./1k	(4) Empl.	(5) Wages	(6) Estabs.
ATT	0.0816***	-0.0180	0.0176*	0.1902**	0.2469	0.1438***
	(0.0274)	(0.1137)	(0.0102)	(0.0743)	(0.2635)	(0.0482)
Pre-treatment mean of dep. var.	1.2007	3.2910	0.5134	4.2928	11.0005	3.1613
ATT as % of pre-treatment mean	6.8%	-0.5%	3.4%	4.4%	2.2%	4.5%
counties	59	59	59	59	59	59

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors.

OA.3 Excluding San Juan: First-Stage and Patient Health

Reports the side-by-side balanced and excluded-San-Juan estimates for the government-hospital birth share, the infant mortality rate, and the low-birth-weight share. The first-stage ownership-share estimate moves from -14.79 to -8.22 percentage points (both $p < 0.01$); the patient-health outcomes are robust to the exclusion.

Table OA.3: Excluding San Juan: Patient-Health and Birth-Share Robustness (Online Appendix)

	Gov. Hosp. Birth Share		IMR		Low BW Rate	
	(1) Bal.	(2) Excl. SJ	(3) Bal.	(4) Excl. SJ	(5) Bal.	(6) Excl. SJ
ATT	-14.79***	-8.22***	-1.205	0.781	0.060	-0.307
	(1.94)	(2.85)	(0.928)	(1.237)	(0.230)	(0.368)
Pre-treatment mean of dep. var.	62.55	62.61	12.620	12.624	9.717	9.712
ATT as % of pre-treatment mean	-23.6%	-13.1%	-9.5%	6.2%	0.6%	-3.2%
counties	78	77	78	77	78	77

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

OA.4 Excluding San Juan: CMS Provider-Stratified

Reports excluded-San-Juan robustness for CMS hospital-bed and services-per-thousand outcomes, stratified by ownership class.

Table OA.4: Excluding San Juan: CMS Provider-Structure Robustness (Online Appendix)

	Beds per 1k pop.		Services per 1k pop.		FTE per 1k pop.	
	(1) Bal.	(2) Excl. SJ	(3) Bal.	(4) Excl. SJ	(5) Bal.	(6) Excl. SJ
ATT	-0.265***	0.090	-0.024**	-0.014	-2.251**	-1.815
	(0.065)	(0.140)	(0.010)	(0.012)	(0.971)	(1.380)
Pre-treatment mean of dep. var.	3.389	3.033	0.487	0.463	10.828	10.288
ATT as % of pre-treatment mean	-7.8%	3.0%	-4.9%	-3.0%	-20.8%	-17.6%
counties	40	39	40	39	33	32

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

OA.5 Spillover-Excluded Sample: First Stage and Patient Health

The first-stage ownership-share estimate on the non-border subsample is -12.14 percentage points ($p < 0.01$, -19.4 percent of pre-mean), economically indistinguishable from the main result; the corresponding patient-health estimates are statistically indistinguishable from zero on this restricted panel.

Table OA.5: Spillover-Excluded Sample: Hospital Birth Share and Patient-Health Outcomes

	Hospital Birth Share (VS)		Patient Health (VS)	
	(1) Gov. Hosp. (%)	(2) Priv. Hosp. (%)	(3) IMR (per 1k)	(4) Low BW (%)
ATT	-12.14***	12.29***	-0.072	0.087
	(2.68)	(2.44)	(1.508)	(0.578)
Pre-treatment mean of dep. var.	62.48	37.00	12.534	9.524
ATT as % of pre-treatment mean	-19.4%	33.2%	-0.6%	0.9%
counties	39	39	39	39

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

OA.6 Spillover-Excluded Sample: CMS Provider Capacity

CMS hospital-bed and services estimates on the spatially-restricted, SUTVA-tightened sample.

Table OA.6: Spillover-Excluded Sample: CMS Provider-Structure Outcomes

	(1) Beds per 1k pop.	(2) Services per 1k pop.	(3) FTE per 1k pop.
ATT	-0.003	-0.020	-3.278
	(0.061)	(0.023)	(2.282)
Pre-treatment mean of dep. var.	3.908	0.657	14.272
ATT as % of pre-treatment mean	-0.1%	-3.1%	-23.0%
N counties	19	19	16

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors.

OA.7 Low-Commuting-Inflow Sample: First Stage and Patient Health

Sample restricted to counties with workplace in-commuting share at or below 0.50 ($n = 64$). The first-stage ownership-share estimate is -7.18 percentage points ($p < 0.05$); the patient-health estimates are robust. Therefore, if we assume that in low-commute counties treatment-effect contamination through cross-county worker flows is minimal, then may assume that our estimates are unbiased, since the effects on the patient-health margin remain consistent with the main estimates.

Table OA.7: Low-Commuting-Inflow Sample: Hospital Birth Share and Patient Health

	(1) Gov. Hosp. Birth Share (%)	(2) IMR (per 1k live births)
ATT	-7.18**	1.327
	(3.29)	(1.392)
Pre-treatment mean of dep. var.	64.11	12.846
ATT as % of pre-treatment mean	-11.2%	10.3%
N counties	64	64

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors. Sample restricted to counties with *low* workplace in-commuting share (in_share ≤ 0.50, where in_share = workers commuting in from other counties divided by total workplace count). The classifier is built from Census 2000 Journey-to-Work tabulations and read from the diagnostic CSV that all_event_studies.R writes at startup.

OA.8 Low-Commuting-Inflow Sample: CMS Provider Capacity

CMS hospital-bed and services estimates on the low-commuting-inflow subsample. CMS supply-side responses in low-commute counties are consistent with main estimates.

Table OA.8: Low-Commuting-Inflow Sample: CMS Provider-Structure Outcomes

	(1) Beds per 1k pop.	(2) Services per 1k pop.	(3) FTE per 1k pop.
ATT	0.087	-0.007	-2.113
	(0.174)	(0.011)	(1.567)
Pre-treatment mean of dep. var.	2.746	0.415	9.730
ATT as % of pre-treatment mean	3.2%	-1.7%	-21.7%
N counties	34	34	27

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors.

OA.9 Heterogeneity by Pre-Reform Population

The infant-mortality estimate in the above-median (larger) cell is -2.564 per thousand live births ($p < 0.01$, -20.8 percent of pre-mean). Second of the four heterogeneity dimensions on which the IMR signal is stronger than the baseline-exposure cut.

Table OA.9: Heterogeneity by Pre-Reform Population (1990–1993 mean) (Online Appendix)

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
Below-median pre-reform population				
ATT	-0.37	-0.154	-1.232	3.657*
	(5.78)	(0.221)	(11.833)	(2.081)
Pre-treatment mean	66.04	1.496	0.050	12.923
ATT as % of pre-mean	-0.6%	-10.3%	-2464.3%	28.3%
N counties	39	20	39	39
Above-median pre-reform population				
ATT	-17.38***	-0.300	-14.651	-2.564***
	(2.27)	(0.228)	(11.447)	(0.921)
Pre-treatment mean	59.33	5.085	1.975	12.347
ATT as % of pre-mean	-29.3%	-5.9%	-741.7%	-20.8%
N counties	39	39	39	39

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors. The classifier is computed once over the strict pre-rollout window (BASELINE_YEARS = 1990–1993) and is therefore time-invariant within county. Counties without a valid baseline value are routed to the below-median bin per the standard censoring rule. *Below-median* = 39 counties; *Above-median* = 39 counties.

OA.10 Heterogeneity by Pre-Reform Healthcare Employment

The infant-mortality estimate in the above-median (higher-pre-reform-healthcare-intensity) cell is -2.633 per thousand live births ($p < 0.01$, -20.7 percent of pre-mean). Third of the four heterogeneity dimensions on which the IMR signal is stronger than the baseline-exposure cut.

Table OA.10: Heterogeneity by Pre-Reform HC Employment (1990–1993 mean)

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
Below-median pre-reform HC employment				
ATT	-4.69	-0.084	-8.171	2.503
	(3.68)	(0.196)	(11.155)	(1.634)
Pre-treatment mean	64.79	1.136	1.440	12.572
ATT as % of pre-mean	-7.2%	-7.4%	-567.5%	19.9%
N counties	48	29	48	48
Above-median pre-reform HC employment				
ATT	-17.87***	-0.433	-7.758	-2.633***
	(2.54)	(0.268)	(13.732)	(0.830)
Pre-treatment mean	58.85	6.816	0.428	12.698
ATT as % of pre-mean	-30.4%	-6.4%	-1810.6%	-20.7%
N counties	30	30	30	30

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Counties are partitioned at the cross-county median of the 1990–1993 mean QCEW Healthcare (NAICS 621+622+623) employment level (annualized from quarterly cells with ≥ 3 valid quarters) (median = 58). The classifier is computed once over the strict pre-rollout window (BASELINE_YEARS = 1990–1993) and is therefore time-invariant within county. Counties without a valid baseline value are routed to the below-median bin. *Below-median* = 48 counties; *Above-median* = 30 counties.

OA.11 Heterogeneity by Workplace Commuting Inflow

The infant-mortality estimate in the above-median (workplace-center) cell is -3.124 per thousand live births ($p < 0.01$, -26.8 percent of pre-mean). Fourth of the four heterogeneity dimensions on which the IMR signal is stronger than the baseline-exposure cut.

Table OA.11: Heterogeneity by Workplace Commuting Inflow (Low vs. High Inflow)

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
Low-inflow workplace counties				
ATT	-7.18**	-0.356	-4.777	1.327
	(3.29)	(0.218)	(9.854)	(1.392)
Pre-treatment mean	64.11	2.910	1.916	12.846
ATT as % of pre-mean	-11.2%	-12.3%	-249.4%	10.3%
N counties	64	45	64	64
High-inflow workplace counties				
ATT	-16.55***	-0.021	7.774	-3.124***
	(3.02)	(0.357)	(11.287)	(1.114)
Pre-treatment mean	55.99	6.974	-2.495	11.671
ATT as % of pre-mean	-29.6%	-0.3%	-311.6%	-26.8%
N counties	14	14	14	14

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Workplace in-commuting share from Census 2000 Journey-to-Work tabulations: in_share = (in-commuters from other counties) / (total workplace count). Threshold cutoff at 0.50 (NOT median). *Low-inflow* = 45 counties (in_share $\leq$ 0.50); *High-inflow* = 14 counties (in_share > 0.50). **Note.** The 'Below-median' and 'Above-median' panels here label the low-inflow and high-inflow groups respectively. **Reading.** High-inflow workplace counties import labor-force flows from neighbors; if treatment effects differ across the inflow margin, that's evidence of cross-county worker spillovers acting on the labor-market response.

OA.12 Per-Thousand-Resident Placebo Industries

C&S estimates on three non-healthcare placebo industries: NAICS 11 (Agriculture, Forestry, and Fishing), NAICS 48–49 (Transportation and Warehousing), and NAICS 71 (Arts, Entertainment, and Recreation). NAICS 31–33 manufacturing is excluded *a priori* as endogenous to the contemporaneous Section 936 corporate-tax phase-out.

Table OA.12: QCEW Placebo Industries: Effects of the reform on Industries Outside the Healthcare Sector

Industry	(1) Empl. per 1k	(2) Wages/Empl per 1k	(3) Wages per 1k	(4) Estabs per 1k
Agriculture, Forestry & Fishing (NAICS 11)				
ATT	0.108	-6.126***	-1,476.474**	0.106
	(0.369)	(1.591)	(741.852)	(0.075)
Pre-treatment mean	12.463	30.797	–	3.463
ATT as % of pre-mean	0.9%	-19.9%	–	3.1%
counties	25	25		25
Transportation & Warehousing (NAICS 48-49)				
ATT	-0.709***	7.571	-688.266***	0.001
	(0.046)	(9.813)	(243.595)	(0.003)
Pre-treatment mean	1.475	240.942	–	0.105
ATT as % of pre-mean	-48.1%	3.1%	–	1.2%
counties	77	77		77
Arts, Entertainment & Recreation (NAICS 71)				
ATT	-0.498	1.994	-553.871	-0.032**
	(0.350)	(1.329)	(558.662)	(0.015)
Pre-treatment mean	2.107	15.207	–	0.155
ATT as % of pre-mean	-23.6%	13.1%	–	-20.6%
counties	9	9		9

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors. **Sample.** QCEW (BLS) county-quarter panel restricted to one of three placebo industries: NAICS 11 (Agriculture, Forestry, Fishing & Hunting); NAICS 48-49 (Transportation & Warehousing); NAICS 71 (Arts, Entertainment & Recreation).

OA.13 Raw-Level Placebo Industries

Raw-level (rather than per-thousand-resident) specifications on the same three placebo industries. Agriculture is null in raw levels, supporting the per-capita-redistribution interpretation of the per-capita Agriculture-wage-bill decline reported in Section 7. Transportation contracts in raw levels by 94.0 percent, consistent with input-output linkage to the phasing-out manufacturing logistics chain. Arts is statistically underpowered. For context, a per-1k placebo null could in principle hide an offsetting population-denominator shock that the per-1k normalization absorbs. Raw-level placebos provide a cleaner test for absolute employment-flow effects.

Table OA.13: QCEW Placebo Industries (Raw Levels): Employment, Wage Bill, and Establishment Counts

Industry	(1) Employment	(2) Wage Bill	(3) Establishments
Agriculture, Forestry & Fishing (NAICS 11)			
ATT	14.5	12,717	0.3
	(13.8)	(20,913)	(2.0)
Pre-treatment mean	293.0	193,903	85.5
ATT as % of pre-mean	4.9%	6.6%	0.3%
N counties	25	25	25
Transportation & Warehousing (NAICS 48-49)			
ATT	-191.9***	-164,320**	1.2***
	(11.8)	(83,737)	(0.3)
Pre-treatment mean	204.1	791,798	7.4
ATT as % of pre-mean	-94.0%	-20.8%	15.9%
N counties	77	77	77
Arts, Entertainment & Recreation (NAICS 71)			
ATT	-63.7	-192,039*	-14.4***
	(51.5)	(106,043)	(3.8)
Pre-treatment mean	309.2	568,279	30.1
ATT as % of pre-mean	-20.6%	-33.8%	-47.7%
N counties	9	9	9

Notes: Three estimators are reported. TWFE: two-way fixed-effects regression with county and year FEs and event-time dummies; the overall ATT is the simple mean of the post-treatment dummy coefficients with delta-method standard error on the post-period. ETWFE: Wooldridge (2021) Extended Two-Way Fixed Effects, with overall ATT as a cohort-size-weighted average. C&S: Callaway and Sant'Anna (2021) doubly-robust ATT, aggregated to a single overall ATT. All three estimators use cluster-robust standard errors at the county level; C&S uses the multiplier bootstrap (1000 replications). Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, based on two-sided tests with cluster-robust standard errors. Raw-level outcomes: Employment is the average quarterly headcount; Wage Bill is the average quarterly real wage bill (deflated; see Section 12b for the deflator); Establishments is the average quarterly establishment count.

OA.14 BRFSS Insurance Coverage by Subgroup

BRFSS subgroup estimates of the ETWFE point estimate on the "has health plan" indicator. The coverage-expansion first stage is positive and significant in five of the six reported subgroups; the 30–44 age group is the one in which the estimate is positive but does not reach conventional significance. The pattern is consistent with the BRFSS analysis covering Cohorts 3–5 (the cohorts for which a pre-treatment BRFSS observation window is available), documented in Appendix 10.

Table OA.14: BRFSS Healthcare Access — Age 18–29

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	-0.0339 (0.0253)	-0.215*** (0.033)	-0.0109 (0.0152)	0.256*** (0.049)
ETWFE	0.0783*** (0.0232)	-0.109* (0.056)	-0.0118 (0.0245)	-0.001 (0.052)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Callaway-Sant'Anna is not run for BRFSS because the dataset's 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents aged 18–29. Aggregated from `all_brfss_coefs.csv` produced by all_event_studies.R; same numerical conventions as Online Appendix Figure OA.15.

Table OA.15: BRFSS Healthcare Access — Age 30–44

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	-0.0053 (0.0145)	0.094*** (0.026)	0.0190** (0.0082)	-0.040 (0.035)
ETWFE	0.0301 (0.0186)	0.025 (0.029)	0.0147 (0.0174)	-0.107** (0.044)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Callaway-Sant'Anna is not run for BRFSS because the dataset's 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents aged 30–44. Aggregated from `all_brfss_coefs.csv` produced by all_event_studies.R; same numerical conventions as Online Appendix Figure OA.16.

Table OA.16: BRFSS Healthcare Access — Age 45–64

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	0.0891*** (0.0084)	0.133*** (0.033)	0.0117 (0.0101)	0.017 (0.040)
ETWFE	0.1052*** (0.0138)	0.100*** (0.032)	-0.0102 (0.0120)	0.102 (0.073)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Callaway-Sant'Anna is not run for BRFSS because the dataset's 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents aged 45–64. Aggregated from `all_brfss_coefs.csv` produced by all_event_studies.R; same numerical conventions as Online Appendix Figure OA.17.

OA.16 Population in Levels and Logarithms

Reports the C&S point estimate on total county population in level and log specifications. The level estimate is +5.1 percent of pre-mean ($p < 0.01$); the log specification is consistent in sign and magnitude. The population growth estimate is the level outcome that complements the demographic-balancing residual (CNMR) reported in Table 6.

Table OA.17: BRFSS Healthcare Access — Age 65 and Older

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	0.0393*** (0.0049)	0.037*** (0.012)	0.0001 (0.0127)	0.138*** (0.033)
ETWFE	0.0280*** (0.0087)	-0.023 (0.014)	-0.0101 (0.0144)	0.211*** (0.052)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Callaway-Sant'Anna is not run for BRFSS because the dataset's 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents aged 65+ (Medicare-eligible). La Reforma did not directly target this group; near-null on health-plan coverage is the expected no-effect benchmark. Aggregated from `all_brfss_coefs.csv` produced by all_event_studies.R; same numerical conventions as Online Appendix Figure OA.18.

Table OA.18: BRFSS Healthcare Access — College Graduates

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	-0.0375*** (0.0104)	0.085*** (0.026)	0.0524*** (0.0103)	0.111*** (0.038)
ETWFE	0.0810*** (0.0212)	0.059* (0.032)	0.0305 (0.0209)	0.201*** (0.059)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Callaway-Sant'Anna is not run for BRFSS because the dataset's 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents with college degree. Aggregated from `all_brfss_coefs.csv` produced by all_event_studies.R; same numerical conventions as Online Appendix Figure OA.20.

Table OA.19: BRFSS Healthcare Access — Employed Respondents

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	-0.0044 (0.0111)	0.029 (15,216.475)	0.0120 (0.0077)	-0.019 (0.034)
ETWFE	0.0460*** (0.0102)	0.016 (0.029)	-0.0233** (0.0104)	-0.144*** (0.037)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Callaway-Sant'Anna is not run for BRFSS because the dataset's 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents reporting employment. Aggregated from `all_brfss_coefs.csv` produced by all_event_studies.R; same numerical conventions as Online Appendix Figure OA.19.

Table OA.20: Effects of The Reform on Log County Population

	(1) Population (levels)	(2) Log Population
ATT	2,528***	0.0357***
	(690)	(0.0086)
Pre-treatment mean of dep. var.	49,898	10.4172
ATT as % of pre-treatment mean	5.1%	0.3%
N counties	78	78

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, based on two-sided tests with cluster-robust standard errors. Total county population (Col. 1) and its natural logarithm (Col. 2).

OA.17 Demographic Reconciliation

Reports the explicit demographic-balancing identity check between the population-level C&S point estimate and the sum of the implied contributions from natural increase (births minus deaths) and net migration. The residual of approximately 66 per thousand reflects (i) measurement error in the CNMR, which is constructed as a residual of three measured flows, and (ii) C&S aggregation weights that differ across panels with different observation counts (the CNMR loses each county's first observation because Pop_{t-1} is unavailable). The two demographic margins are reported jointly because the disagreement carries identification content.

Table OA.21: Demographic Margins of Adjustment: Reconciling Levels and Flows

	Population (Levels)		Net Migration	Vital Rates (Flows)	
	(1) Population	(2) Log Population	(3) CNMR (per 1k)	(4) Crude Birth Rate (per 1k)	(5) Crude Death Rate (per 1k)
ATT	2,528***	0.0357***	-14.651**	-0.564**	0.047
	(690)	(0.0086)	(7.389)	(0.225)	(0.114)
Pre-treatment mean of dep. var.	49,898	10.4172	1.059	18.267	7.293
ATT as % of pre-treatment mean	5.1%	0.3%	-1382.9%	-3.1%	0.6%
N counties	78	78	78	78	78

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, based on two-sided tests with cluster-robust standard errors. The demographic identity for any closed area between two periods is $\Delta\text{Population}_{t,t-1} = (\text{Births}_t - \text{Deaths}_t) + \text{Net Migration}_t$, and dividing by average population yields the per-1k version: $\Delta \log P \approx \text{CBR} - \text{CDR} + \text{CNMR}$. Columns (3)–(5) report the C&S ATT for each of the three flow components on the right-hand side; columns (1)–(2) report the left-hand side in absolute and proportional units.

OA.18 NVSS Individual-Level Birth-Outcome Robustness

Reports C&S estimates on individual-level NVSS natality records using the conception-year-treatment indicator. Outcomes: birth weight, the count of prenatal visits, the trimester of prenatal-care initiation, and the five-minute Apgar score. The point estimates are statistically indistinguishable from zero, consistent with the county-aggregate Vital Statistics null reported in Table 5.

Table OA.22: NVSS Individual-Level Birth Outcomes — Effects of The Reform

	(1) Birth Weight (g)	(2) Prenatal Visits	(3) Apgar (5-min)	(4) Pr(LBW)	(5) Pr(VLBW)
ATT (C&S)	-0.75	-0.18	0.002	0.0015	-0.0001
	(9.99)	(0.17)	(0.020)	(0.0036)	(0.0013)

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county (municipio) level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, based on two-sided tests with cluster-robust standard errors. The variables are continuous birth-weight, prenatal-visits, and Apgar measures plus binary low/very-low-birth-weight indicators.

OA.19 Heterogeneity by Pre-Reform Poverty Rate

Partitions the seventy-eight counties at the median Census 2000 poverty rate (U.S. Census Bureau, 2000). The healthcare-employment estimate in the above-median-poverty cell is -31.2 percent of pre-mean ($p < 0.10$); the result is directionally consistent with concentration of the contraction in slack-labor-market cells. Sample-split power limits formal inference on the access-component-versus-ownership-component decomposition contemplated by the framework of Section 4.

Table OA.23: Heterogeneity by Census 2000 Poverty Rate

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
Below-median poverty rate				
ATT	-17.81***	-0.295	-15.773	-1.622*
	(1.93)	(0.214)	(11.426)	(0.984)
Pre-treatment mean	57.92	5.481	2.166	11.970
ATT as % of pre-mean	-30.7%	-5.4%	-728.1%	-13.5%
N counties	39	34	39	39
Above-median poverty rate				
ATT	-2.49	-0.557*	1.249	2.066
	(4.76)	(0.290)	(11.907)	(1.972)
Pre-treatment mean	67.14	1.787	-0.037	13.265
ATT as % of pre-mean	-3.7%	-31.2%	-3378.7%	15.6%
N counties	39	25	39	39

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Counties are partitioned at the cross-county median of the Census 2000 % of residents below the federal poverty line (median = 54.55%). The classifier is computed once over the strict pre-rollout window (baseline years = 1990–1993) and is therefore time-invariant within county. Counties without a valid baseline value are routed to the below-median bin per the standard censoring rule. *Below-median* = 39 counties; *Above-median* = 39 counties. Census 2000 is partially post-rollout for cohorts 1–4 (1994–1998) and concurrent with cohort 5 (2000), so the classifier is partially endogenous to the reform itself; yet we retain it because no comparable pre-reform county-level poverty series exists.

OA.20 Heterogeneity by Pre-Reform Unemployment Rate

Partitions the seventy-eight counties at the median pre-reform unemployment rate. The healthcare-employment estimate in the above-median-unemployment cell is -17.8 percent of pre-mean (statistically insignificant); the directional pattern is consistent with the poverty-rate cut.

Table OA.24: Heterogeneity by Pre-Reform Unemployment Rate (BLS, 1990–1993 mean)

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
Below-median pre-reform unemployment rate				
ATT	-14.23***	-0.331	-5.217	-2.617**
	(2.24)	(0.203)	(10.675)	(1.146)
Pre-treatment mean	58.72	5.027	0.059	12.332
ATT as % of pre-mean	-24.2%	-6.6%	-8888.1%	-21.2%
N counties	39	32	39	39
Above-median pre-reform unemployment rate				
ATT	-7.37***	-0.454	-11.417	3.213
	(2.13)	(0.358)	(11.903)	(2.294)
Pre-treatment mean	66.50	2.550	2.100	12.912
ATT as % of pre-mean	-11.1%	-17.8%	-543.7%	24.9%
N counties	39	27	39	39

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: $^{*}\, p < 0.10$, $^{**}\, p < 0.05$, $^{***}\, p < 0.01$, based on two-sided tests with cluster-robust standard errors. Counties are partitioned at the cross-county median of the 1990–1993 mean PR DOL BLS-standard unemployment rate (unemployed / labor force $\times$ 100) (median = 18.62%). The classifier is computed once over the strict pre-rollout window (baseline years = 1990–1993) and is therefore time-invariant within county. counties without a valid baseline value are routed to the below-median bin. *Below-median* = 39 counties; *Above-median* = 39 counties.

OA.21 Adult Cause-Specific Mortality

Reports C&S estimates on adult cause-specific mortality outcomes (heart disease, cerebrovascular disease, diabetes, accidents, homicide). Several causes of death - such as cardiovascular disease, cerebrovascular disease, diabetes, and pneumonia or influenza- have biologically plausible response horizons that extend well beyond our four-year post-reform observation window. Conversely, acute causes like accidents and homicides likely reflect the true, unbiased mortality rate; because these events demand immediate attention from local providers, they are less susceptible to the confounding spillover effects of healthcare access in neighboring counties. Similarly, maternal mortality operates on a longer time horizon. Preventing maternal death depends heavily on long-term, systemic healthcare managed over decades, and is increasingly confounded by demographic shifts, such as women entering pregnancy later in life. Consequently, we select the Infant Mortality Rate (IMR) as our primary healthcare indicator. Unlike maternal health, saving a newborn relies predominantly on immediate, high-tech medical interventions, making IMR highly sensitive to short-term, localized healthcare quality. The remaining mortality estimates are reported solely for completeness and should not be interpreted as identified treatment effects.

Table OA.25: Cause-Specific Mortality Effects of The Reform

	(1) Heart	(2) Cerebro.	(3) Diabetes	(4) Pneum./Flu	(5) Perinatal	(6) Maternal	(7) Accidents	(8) Homicide
ATT	-0.096**	-0.041**	-0.010	0.009	–	0.002*	0.138	0.107***
	(0.046)	(0.020)	(0.034)	(0.021)		(0.001)	(0.101)	(0.015)
Pre-treatment mean of dep. var.	1.572	0.370	0.524	0.298	–	0.003	0.344	0.143
ATT as % of pre-treatment mean	-6.1%	-11.0%	-2.0%	3.1%	–	73.1%	40.1%	75.0%
N counties	78	78	78	78	–	78	78	78

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. **Outcomes.** Cause-specific death rates per 1,000 residents. **Caveat.**